

FLUX 系列模型与超分辨率技术深度洞察报告

FLUX.1 / FLUX.2 · DiT · Rectified Flow · Flow Matching · Super-Resolution

版本：2026-08-05 证据化学习手册初稿 | 资料截止：2026-08-03 | 研究对象：19 篇已下载并通过 PDF 校验的论文/技术报告

本报告采用 ‘主线教材 + 全量论文图谱’ 方案。中文负责解释技术逻辑，英文保留关键术语、公式和指标名称；原文截图均标注论文 Figure/Table 与 PDF 页码。论文正文没有披露的内容，不以推测替代。

阅读说明与证据边界 | Reading protocol

- 正式 FLUX 论文：正文明确使用 FLUX.1/FLUX.2 作为 backbone、teacher、adaptation target、实验对象或主要评测模型。
- 基础理论论文：只作为 DiT、Rectified Flow、Flow Matching、MMDiT 的教材，不冒充 FLUX 官方训练细节。
- 性能数字：仅记录论文明确报告的 resolution、NFE、latency、speedup、GPU、PSNR/SSIM/LPIPS/CLIP-IQA/ImageReward 等。
- 业务判断：属于基于论文证据的工程推论，会与论文直接事实分开表述。

一页结论 | Executive synthesis

FLUX 相关超分研究正在形成一条清晰路线：先把大规模 flow-matching DiT 当作生成先验，再通过 distillation、LoRA、ControlNet、preference optimization 或 inference-time scaling 把先验转化为 restoration capability。研究重点已经从 ‘能否生成高清图’ 转向 ‘如何在一步/少步、无分块、Profile 可控和有限算力下生成可信细节’。

最重要的矛盾是 fidelity-perception trade-off：PSNR/SSIM 追求与输入/GT 一致，扩散先验则倾向于补全语义上合理但未被观测到的纹理。FoA-SR 将这种冲突显式化为 Faithful/Aesthetic 两种 LoRA；FluxSR、DreamSR、OP4KSR 则分别从轨迹蒸馏、感受野和周期伪影三个方向压缩风险。

从部署角度，真正可落地的组合不是单一 ‘更大模型’，而是：FLUX prior + 任务适配 + one-step/少步 distillation + cache/sparse/spatial acceleration + profile-aware quality gate。云端相册和数字变焦最先受益，端侧还需独立完成量化、kernel、内存和热稳定性验证。

技术演进时间轴 | Technology timeline

时间	代表路线	技术变化
2022	DiT	Transformer 作为 diffusion backbone, 建立 scaling 视角
2022-2023	Rectified Flow / Flow Matching	从 noise 到 data 的 velocity field 与更直的生成轨迹
2024	SD3 / MMDiT	双流 text/image token 与联合 attention, 面向高分辨率生成
2024-2025	FLUX.1 / Kontext	大规模 flow-matching multimodal latent transformer 与 in-context editing
2025	FluxSR / FluxIR / LucidFlux	生成先验被适配为 real-world SR 与 image restoration
2026	DreamSR / OP4KSR / FoA-SR	4K、patch-free、周期伪影与 profile-aware preference
2024-2026	Quantization / Cache / Sparse / RALU	把 FLUX 的计算图拆解到权重、时间、空间和 kernel 层面

第一篇 基础理论教材 | Foundations

这一篇回答 ‘为什么 FLUX 会采用 DiT + Rectified Flow + Flow Matching + multimodal transformer’ 。基础论文的价值在于建立概念坐标系；不要把它们 ImageNet 生成结果直接和 real-world SR 的 PSNR 进行横向排名。

DiT：把扩散模型的主干从 U-Net 迁移到 Transformer

论文角色	理论基础，不直接使用 FLUX
作者/机构	William Peebles; Saining Xie / UC Berkeley / NYU; Meta AI internship
发表状态	ICCV 2023; arXiv:2212.09748
纳入等级	基础教材

研究问题与前序缺陷 | Research question

传统扩散模型长期依赖 U-Net。DiT 追问的是：如果把 latent diffusion 的去噪主干替换为 Transformer，模型容量、token 数和计算量是否能形成可预测的 scaling law?

方法拆解 | Method

输入 latent 被切分为 patch tokens，加入位置编码后进入 Transformer blocks；条件通过 adaLN-Zero 注入。与卷积网络相比，DiT 的计算规模更容易通过 depth、width、token 数和 Gflops 控制。

关键公式 | Key equation: $x_t = \alpha_t x_0 + \sigma_t \epsilon$; $h' = h + \text{adaLN-Zero}(h, c)$

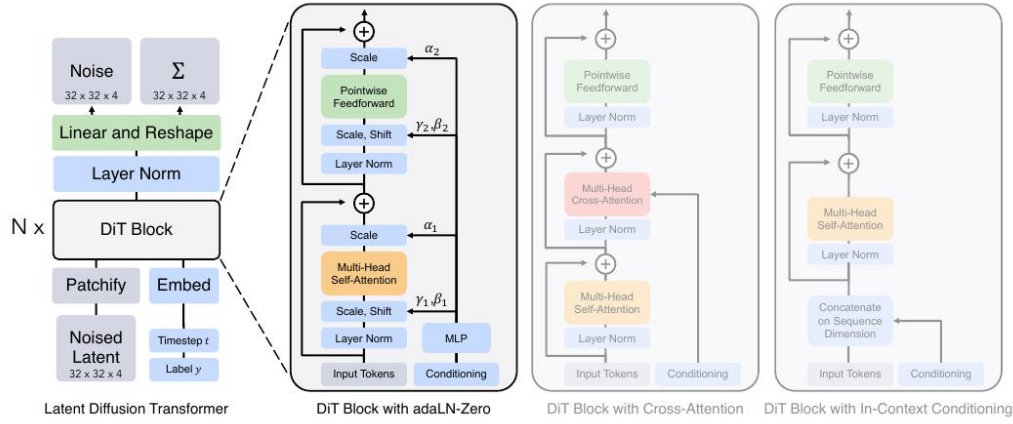


Figure 3. **The Diffusion Transformer (DiT) architecture.** *Left:* We train conditional latent DiT models. The input latent is decomposed into patches and processed by several DiT blocks. *Right:* Details of our DiT blocks. We experiment with variants of standard transformer blocks that incorporate conditioning via adaptive layer norm, cross-attention and extra input tokens. Adaptive layer norm works best.

Denoising diffusion probabilistic models (DDPMs).

Diffusion [19, 54] and score-based generative models [22, 56] have been particularly successful as generative models of images [35, 46, 48, 50], in many cases outperforming generative adversarial networks (GANs) [12] which had previously been state-of-the-art. Improvements in DDPMs over the past two years have largely been driven by improved sampling techniques [19, 27, 55], most notably classifier-free guidance [21], reformulating diffusion models to predict noise instead of pixels [19] and using cascaded DDPM pipelines where low-resolution base diffusion models are trained in parallel with upsamplers [9, 20]. For all the diffusion models listed above, convolutional U-Nets [49] are the de-facto choice of backbone architecture. Concurrent work [24] introduced a novel, efficient architecture based on attention for DDPMs; we explore pure transformers.

Architecture complexity. When evaluating architecture complexity in the image generation literature, it is fairly common practice to use parameter counts. In general, parameter counts can be poor proxies for the complexity of image models since they do not account for, e.g., image resolution which significantly impacts performance [44, 45]. Instead, much of the model complexity analysis in this paper is through the lens of theoretical Gflops. This brings us in-line with the architecture design literature where Gflops are widely-used to gauge complexity. In practice, the golden complexity metric is still up for debate as it frequently depends on particular application scenarios. Nichol and Dhariwal’s seminal work improving diffusion models [9, 36] is most related to us—there, they analyzed the scalability and Gflop properties of the U-Net architecture class. In this paper, we focus on the transformer class.

3. Diffusion Transformers

3.1. Preliminaries

Diffusion formulation. Before introducing our architecture, we briefly review some basic concepts needed to understand diffusion models (DDPMs) [19, 54]. Gaussian diffusion models assume a forward noising process which gradually applies noise to real data x_0 : $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1 - \alpha_t)\mathbf{I})$, where constants α_t are hyperparameters. By applying the reparameterization trick, we can sample $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$.

Diffusion models are trained to learn the reverse process that inverts forward process corruptions: $p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t), \Sigma_\theta(x_t))$, where neural networks are used to predict the statistics of p_θ . The reverse process model is trained with the variational lower bound [30] of the log-likelihood of x_0 , which reduces to $\mathcal{L}(\theta) = -p(x_0|x_1) + \sum_t \mathcal{D}_{KL}(q^*(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t))$, excluding an additional term irrelevant for training. Since both q^* and p_θ are Gaussian, $\mathcal{D}_{KL}$ can be evaluated with the mean and covariance of the two distributions. By reparameterizing μ_θ as a noise prediction network ϵ_θ , the model can be trained using simple mean-squared error between the predicted noise $\epsilon_\theta(x_t)$ and the ground truth sampled Gaussian noise ϵ_t : $\mathcal{L}_{simple}(\theta) = \|\epsilon_\theta(x_t) - \epsilon_t\|_2^2$. But, in order to train diffusion models with a learned reverse process covariance Σ_θ , the full $\mathcal{D}_{KL}$ term needs to be optimized. We follow Nichol and Dhariwal’s approach [36]: train ϵ_θ with $\mathcal{L}_{simple}$, and train Σ_θ with the full $\mathcal{L}$. Once p_θ is trained, new images can be sampled by initializing $x_{t_{max}} \sim \mathcal{N}(0, \mathbf{I})$ and sampling $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$ via the reparameterization trick.

Scalable Diffusion Models with Transformers

William Peebles*
UC Berkeley

Saining Xie
New York University



Figure 1. **Diffusion models with transformer backbones achieve state-of-the-art image quality.** We show selected samples from two of our class-conditional DiT-XL/2 models trained on ImageNet at 512×512 and 256×256 resolution, respectively.

Abstract

We explore a new class of diffusion models based on the transformer architecture. We train latent diffusion models of images, replacing the commonly-used U-Net backbone with a transformer that operates on latent patches. We analyze the scalability of our Diffusion Transformers (DiTs) through the lens of forward pass complexity as measured by Gflops. We find that DiTs with higher Gflops—through increased transformer depth/width or increased number of input tokens—consistently have lower FID. In addition to possessing good scalability properties, our largest DiT-XL/2 models outperform all prior diffusion models on the class-conditional ImageNet 512×512 and 256×256 benchmarks, achieving a state-of-the-art FID of 2.27 on the latter.

1. Introduction

Machine learning is experiencing a renaissance powered by transformers. Over the past five years, neural architectures for natural language processing [8, 42], vision [10] and several other domains have largely been subsumed by transformers [60]. Many classes of image-level generative models remain holdouts to the trend, though—while transformers see widespread use in autoregressive models [3, 6, 43, 47], they have seen less adoption in other generative modeling frameworks. For example, diffusion models have been at the forefront of recent advances in image-level generative models [9, 46]; yet, they all adopt a convolutional U-Net architecture as the de-facto choice of backbone.

* Work done during an internship at Meta AI, FAIR Team.
Code and project page available [here](#).

原文截图: DiT: 把扩散模型的主干从 U-Net 迁移到 Transformer, Figure 1, generated samples; 来源 PDF 第 1 页。

创新点与真正解决的问题 | Contribution

真正的贡献不是提出一个更复杂的模块，而是建立了可扩展、可测量的 Transformer 扩散主干。它为后续 MMDiT/FLUX 采用双流与单流 Transformer 提供了工程母体。

它真正解决的问题是：解决了扩散主干难以按模型规模稳定扩展的问题，并证明更高 Gflops 通常对应更低 FID。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

实验主体是 ImageNet class-conditional generation，不等同于真实图像复原；其 adaLN 条件接口也不能直接代表 FLUX 的文本/图像双模态注入。

对相机业务的启示是：Transformer 主干能够把全局结构建模与高分辨率 token 处理统一起来，但 token 数带来的二次复杂度会直接制约端侧实时性。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

Rectified Flow：把生成轨迹变直，以减少采样步数

论文角色	理论基础，不直接使用 FLUX
作者/机构	Xingchao Liu et al. / University of Texas at Austin and collaborators
发表状态	ICLR 2023; arXiv:2209.03003
纳入等级	基础教材

研究问题与前序缺陷 | Research question

扩散/ODE 生成通常需要沿弯曲的概率路径积分，少步数离散会产生较大误差。Rectified Flow 的核心问题是如何让从噪声到数据的运输路径更接近直线。

方法拆解 | Method

先用线性插值构造条件路径，再回归其速度场。通过 reflow 反复重配对，可以把原始耦合变成更直的流，从而允许更粗的 Euler 离散。

关键公式 | Key equation: $Z_t = (1-t)X_0 + tX_1$; $v_t(z) = E[X_1 - X_0 | Z_t = z]$; $dZ_t/dt = v_t(Z_t)$

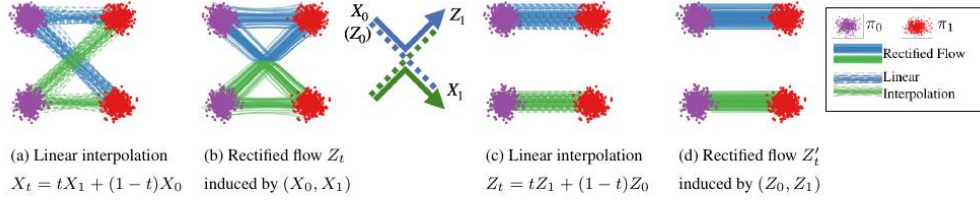


Figure 2: (a) Linear interpolation of data input $(X_0, X_1) \sim \pi_0 \times \pi_1$. (b) The rectified flow Z_t induced by (X_0, X_1) ; the trajectories are “rewired” at the intersection points to avoid the crossing. (c) The linear interpolation of the end points (Z_0, Z_1) of flow Z_t . (d) The rectified flow induced from (Z_0, Z_1) , which follows straight paths.

Empirically, rectified flow can yield high-quality results for image generation when simulated with a very few number of Euler steps (see Figure 1, top row). Moreover, with just one step of reflow, the flow becomes nearly straight and hence yield good results with a single Euler discretization step (Figure 1, the second row). This substantially improves over the standard denoising diffusion methods. Quantitatively, we claim a state-of-the-art result of FID (4.85) and recall (0.51) on CIFAR10 for one-step fast diffusion/flow models [5, 48, 91, 99, 47]. The same algorithm also achieves superb result on domain transfer tasks such as image-to-image translation (see the bottom two rows of Figure 1) and transfer learning.

2 Method

We provide a quick overview of the method in Section 2.1, followed with some discussion and remarks in Section 2.2. We introduce a nonlinear extension of our method in Section 2.3, with which we clarify the connection and advantages of our method with the method of probability flow ODEs [73] and DDIM [70].

2.1 Overview

Rectified flow Given empirical observations of $X_0 \sim \pi_0, X_1 \sim \pi_1$, the rectified flow induced from (X_0, X_1) is an ordinary differentiable model (ODE) on time $t \in [0, 1]$,

$$dZ_t = v(Z_t, t)dt,$$

which converts Z_0 from π_0 to a Z_1 following π_1 . The drift force $v: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is set to drive the flow to follow the direction $(X_1 - X_0)$ of the linear path pointing from X_0 to X_1 as much as possible, by solving a simple least squares regression problem:

$$\min_v \int_0^1 \mathbb{E} \left[\|(X_1 - X_0) - v(X_t, t)\|^2 \right] dt, \quad \text{with } X_t = tX_1 + (1-t)X_0, \quad (1)$$

where X_t is the linear interpolation of X_0 and X_1 . Naively, X_t follows the ODE of $dX_t = (X_1 - X_0)dt$, which is non-causal (or anticipating) as the update of X_t requires the information of the final point X_1 . By fitting the drift v with $X_1 - X_0$, the rectified flow *causalizes* the paths of linear interpolation X_t , yielding an ODE flow that can be simulated without seeing the future.

In practice, we parameterize v with a neural network or other nonlinear models and solve (1) with any off-the-shelf stochastic optimizer, such as stochastic gradient descent, with empirical draws of (X_0, X_1) . See

创新点与真正解决的问题 | Contribution

把少步采样从经验性的 scheduler 调参，转化为轨迹几何问题：路径越直，有限步积分误差越小。FLUX 的 flow matching 体系直接继承这一思想。

它真正解决的问题是：解决了流模型在低 NFE 下轨迹弯曲、误差累积和采样效率不足的问题。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

直线化不保证所有数据分布都能用极少步数无损表达；重流化有额外训练成本，且理论 toy case 与 12B 级文本图像模型之间存在尺度鸿沟。

真实相机精修若要落地，少步采样可降低功耗和交互延迟，但还必须联合研究结构保真、噪声建模和细节幻觉。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

Flow Matching：直接学习概率路径上的速度场

论文角色	理论基础，不直接使用 FLUX
作者/机构	Yaron Lipman et al. / Meta AI (FAIR) / Weizmann Institute
发表状态	ICLR 2023; arXiv:2210.02747
纳入等级	基础教材

研究问题与前序缺陷 | Research question

连续归一化流的最大似然训练需要昂贵的 ODE 模拟，而扩散模型的 score matching 又限制了路径选择。Flow Matching 希望用 simulation-free 的回归目标训练向量场。

方法拆解 | Method

从 source noise 和 target data 构造 conditional probability path，直接回归该路径的条件速度。Optimal Transport path 通过更短、更有序的运输连接改善训练和采样。

关键公式 | Key equation: $L_{FM}(\theta) = E_{\{t, p_t(x|x_1), p_{data}(x_1)\}} \|v_\theta(t, x) - u_t(x|x_1)\|^2;$

$$dx_t/dt = v_\theta(t, x_t)$$

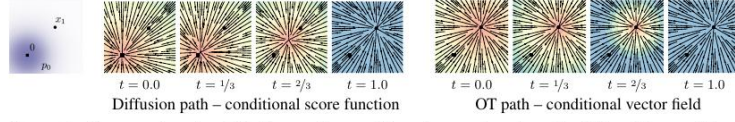


Figure 2: Compared to the diffusion path’s conditional score function, the OT path’s conditional vector field has constant direction in time and is arguably simpler to fit with a parametric model. Note the blue color denotes larger magnitude while red color denotes smaller magnitude.

which, in contrast to the diffusion conditional VF (equation 19), is defined for all $t \in [0, 1]$. The conditional flow that corresponds to $u_t(x|x_1)$ is

$$\psi_t(x) = (1 - (1 - \sigma_{\min})t)x + tx_1, \quad (22)$$

and in this case, the CFM loss (see equations 9, 14) takes the form:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, q(x_1), p(x_0)} \left\| v_t(\psi_t(x_0)) - (x_1 - (1 - \sigma_{\min})x_0) \right\|^2. \quad (23)$$

Allowing the mean and std to change linearly not only leads to simple and intuitive paths, but it is actually also optimal in the following sense. The conditional flow $\psi_t(x)$ is in fact the Optimal Transport (OT) *displacement map* between the two Gaussians $p_0(x|x_1)$ and $p_1(x|x_1)$. The OT *interpolant*, which is a probability path, is defined to be (see Definition 1.1 in McCann (1997)):

$$p_t = [(1 - t)\text{id} + t\psi]_* p_0 \quad (24)$$

where $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the OT map pushing p_0 to p_1 , id denotes the identity map, *i.e.*, $\text{id}(x) = x$, and $(1 - t)\text{id} + t\psi$ is called the OT displacement map. Example 1.7 in McCann (1997) shows, that in our case of two Gaussians where the first is a standard one, the OT displacement map takes the form of equation 22.

Intuitively, particles under the OT displacement map always move in straight line trajectories and with constant speed. Figure 3 depicts sampling paths for the diffusion and OT conditional VFs. Interestingly, we find that sampling trajectory from diffusion paths can “overshoot” the final sample, resulting in unnecessary backtracking, whilst the OT paths are guaranteed to stay straight.

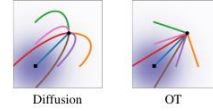


Figure 3: Diffusion and OT trajectories.

Figure 2 compares the diffusion conditional score function (the regression target in a typical diffusion methods), *i.e.*, $\nabla \log p_t(x|x_1)$ with p_t defined as in equation 18, with the OT conditional VF (equation 21). The start (p_0) and end (p_1) Gaussians are identical in both examples. An interesting observation is that the OT VF has a constant direction in time, which arguably leads to a simpler regression task. This property can also be verified directly from equation 21 as the VF can be written in the form $u_t(x|x_1) = g(t)h(x|x_1)$. Figure 8 in the Appendix shows a visualization of the Diffusion VF. Lastly, we note that although the conditional flow is optimal, this by no means imply that the marginal VF is an optimal transport solution. Nevertheless, we expect the marginal vector field to remain relatively simple.

5 RELATED WORK

Continuous Normalizing Flows were introduced in (Chen et al., 2018) as a continuous-time version of Normalizing Flows (see *e.g.*, Kobyzev et al. (2020); Papamakarios et al. (2021) for an overview). Originally, CNFs are trained with the maximum likelihood objective, but this involves expensive ODE simulations for the forward and backward propagation, resulting in high time complexity due to the sequential nature of ODE simulations. Although some works demonstrated the capability of CNF generative models for image synthesis (Grathwohl et al., 2018), scaling up to very high dimensional images is inherently difficult. A number of works attempted to regularize the ODE to be easier to solve, *e.g.*, using augmentation (Dupont et al., 2019), adding regularization terms (Yang & Karniadakis, 2019; Finlay et al., 2020; Onken et al., 2021; Tong et al., 2020; Kelly et al., 2020), or stochastically sampling the integration interval (Du et al., 2022). These works merely aim to regularize the ODE but do not change the fundamental training algorithm.

原文截图: Flow Matching: 直接学习概率路径上的速度场, Figures 2-3, OT and diffusion probability paths; 来源 PDF 第 6 页。

创新点与真正解决的问题 | Contribution

将训练目标从 ‘预测噪声’ 换成 ‘预测速度’，把路径设计、ODE solver 和模型训练解耦。FLUX.1/2 的 rectified-flow transformer 正是这一范式在大规模图像生成上的产品化延伸。

它真正解决的问题是：解决了大规模 CNF 训练的模拟成本和路径限制问题，并为少步数确定性采样提供理论接口。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

论文主要验证 ImageNet 生成质量与 likelihood；并未回答真实低质图像如何映射到生成路径上的逆问题。

相机超分可以把低清输入作为条件端点或 latent anchor，利用速度场完成可控细节生成，但必须引入 fidelity loss 或 profile control。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

OT-CFM：用 minibatch Optimal Transport 改善路径耦合

论文角色	支撑理论，不直接使用 FLUX
作者/机构	文献基础模块 / Flow Matching supporting work
发表状态	arXiv:2302.00482
纳入等级	支撑理论

研究问题与前序缺陷 | Research question

随机配对的 source-target 样本可能诱导较长、较弯的条件路径，影响 vector-field 回归效率。

方法拆解 | Method

在 minibatch 内求近似最优传输匹配，以较短的 displacement interpolation 形成 conditional flow matching 数据。

关键公式 | Key equation: $\pi^* = \operatorname{argmin}_{\pi} \mathbb{E}_{\{(x_0, x_1) \sim \pi\}} \|x_1 - x_0\|^2$; $x_t = (1-t)x_0 + tx_1$

Published in Transactions on Machine Learning Research (03/2024)

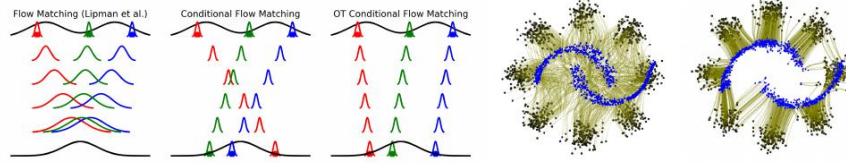


Figure 1: **Left:** Conditional flows from FM (Lipman et al., 2023), I-CFM (§3.2.2), and OT-CFM (§3.2.3). **Right:** Learned flows (green) from moons (blue) to 8 Gaussians (black) using I-CFM (centre-right) and OT-CFM (far right).

and Schrödinger bridge problems. For high-dimensional image generation, we also propose improved and reproducible training practices for flow-based models that significantly improve the performance of algorithms from past work (§5).

- (4) We release a Python package, `torchcfm`, that unifies new and existing algorithms for training flow-based generative models under a shared interface and provides implementations of our main experiments. The Python code is available at <https://github.com/atong01/conditional-flow-matching>

2 Background: Optimal transport and neural ODEs

Throughout the paper, we consider the setting of a pair of data distributions over $\mathbb{R}^d$ with (possibly unknown) densities $q(x_0)$ and $q(x_1)$ (also denoted q_0, q_1). Generative modeling considers the task of fitting a mapping f from $\mathbb{R}^d$ to $\mathbb{R}^d$ that transforms q_0 to q_1 , that is, if x_0 is distributed with density q_0 then $f(x_0)$ is distributed with density q_1 . This includes both the typical case when q_0 is an easily sampled density, such as a Gaussian, and the case when q_0 and q_1 are empirical data distributions available as finite sets of samples.

2.1 ODEs and probability flows

A smooth¹ time-varying vector field $u : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ defines an ordinary differential equation:

$$dx = u_t(x) dt, \quad (1)$$

where we use the notation $u_t(x)$ interchangeably with $u(t, x)$. Denote by $\phi_t(x)$ the solution of the ODE (1) with initial condition $\phi_0(x) = x$; that is, $\phi_t(x)$ is the point x transported along the vector field u from time 0 up to time t .

Given a density p_0 over $\mathbb{R}^d$, the integration map ϕ_t induces a pushforward $p_t := [\phi_t]_{\#}(p_0)$, which is the density of points $x \sim p_0$ transported along u from time 0 to time t . The time-varying density p_t , viewed as a function $p : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}$, is characterized by the well-known *continuity equation*:

$$\frac{\partial p}{\partial t} = -\nabla \cdot (p_t u_t) \quad (2)$$

and the initial conditions p_0 . Under these conditions, u is said to be a *probability flow ODE* for p , and p is the *(marginal) probability path*² generated by u .

Approximating ODEs with neural networks. Suppose the probability path $p_t(x)$ and the vector field $u_t(x)$ generating it are known and $p_t(x)$ can be tractably sampled. If $v_{\theta}(\cdot, \cdot) : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a time-dependent vector field parametrized as a neural network with weights θ , v_{θ} can be regressed to u via

¹To be precise, to ensure the uniqueness of integral curves (and thus of the corresponding flow), we assume the vector field u is at least locally Lipschitz in x and Bochner integrable in t .

²The terminology is due to $t \mapsto p_t$ being a path on the infinite-dimensional manifold of probability distributions on $\mathbb{R}^d$.

创新点与真正解决的问题 | Contribution

把 OT 的配对结构嵌入小批量训练，不需要完整数据集上的全局最优传输。

它真正解决的问题是：减少不必要的路径弯曲，改善 FM 目标的统计效率。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

它是理论支撑，不应被误写成 FLUX 的官方结构或训练细节。

对 SR 的启示是：低清/高质样本和噪声/图像样本的配对方式，会影响模型学习到的修复轨迹。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

Stable Diffusion 3 / MMDiT：双流到单流的多模态 Transformer

论文角色	FLUX 结构前置，不等同于 FLUX 官方论文
作者/机构	Stable Diffusion 3 technical report authors / Stability AI and collaborators
发表状态	arXiv:2403.03206
纳入等级	关键前置

研究问题与前序缺陷 | Research question

文本图像生成需要同时保持语言理解和图像空间建模。单一 cross-attention 容易造成模态间信息瓶颈，尤其在文字渲染与复杂构图上。

方法拆解 | Method

MMDiT 使用 image stream 与 text stream 的独立参数投影，在 attention 中联合交互；随后通过 single-stream blocks 融合多模态信息。RMSNorm、QK-normalization 与 rectified flow 共同稳定高分辨率训练。

关键公式 | Key equation: $Q_i = X_i W^Q$, $K_i = X_i W^K$, $V_i = X_i W^V$;

$\text{Attn}(X) = \text{softmax}(QK^T / \sqrt{d})V$

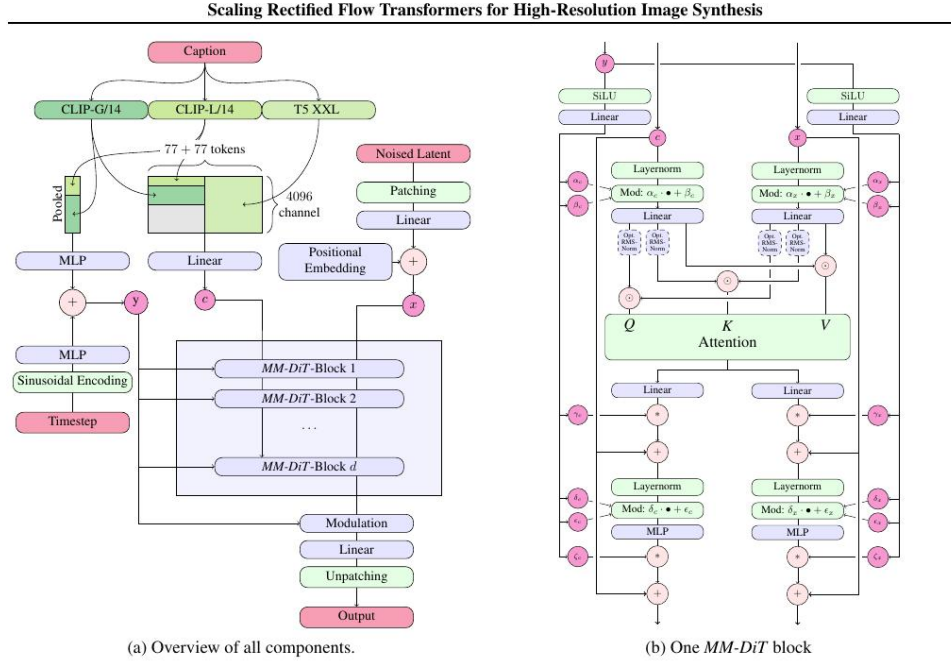


Figure 2. **Our model architecture.** Concatenation is indicated by $\odot$ and element-wise multiplication by $*$. The RMS-Norm for Q and K can be added to stabilize training runs. Best viewed zoomed in.

concatenate the two sequences. We then follow DiT and apply a sequence of modulated attention and MLPs.

Since text and image embeddings are conceptually quite different, we use two separate sets of weights for the two modalities. As shown in Figure 2b, this is equivalent to having two independent transformers for each modality, but joining the sequences of the two modalities for the attention operation, such that both representations can work in their own space yet take the other one into account.

For our scaling experiments, we parameterize the size of the model in terms of the model’s depth d , i.e. the number of attention blocks, by setting the hidden size to $64 \cdot d$ (expanded to $4 \cdot 64 \cdot d$ channels in the MLP blocks), and the number of attention heads equal to d .

5. Experiments

5.1. Improving Rectified Flows

We aim to understand which of the approaches for simulation-free training of normalizing flows as in Equation 1 is the most efficient. To enable comparisons across different approaches, we control for the optimization algorithm, the model architecture, the dataset and samplers. In

addition, the losses of different approaches are incomparable and also do not necessarily correlate with the quality of output samples; hence we need evaluation metrics that allow for a comparison between approaches. We train models on ImageNet (Russakovsky et al., 2014) and CC12M (Changpinyo et al., 2021), and evaluate both the training and the EMA weights of the models during training using validation losses, CLIP scores (Radford et al., 2021; Hessel et al., 2021), and FID (Heusel et al., 2017) under different sampler settings (different guidance scales and sampling steps). We calculate the FID on CLIP features as proposed by (Sauer et al., 2021). All metrics are evaluated on the COCO-2014 validation split (Lin et al., 2014). Full details on the training and sampling hyperparameters are provided in Appendix B.3.

5.1.1. RESULTS

We train each of 61 different formulations on the two datasets. We include the following variants from Section 3:

- Both ϵ - and v -prediction loss with linear (eps/linear , v/linear) and cosine (eps/cos , v/cos) schedule.
- RF loss with $\pi_{\text{mode}}(t; s)$ ($\text{rf}/\text{mode}(s)$) with 7 values for s chosen uniformly between -1 and 1.75 , and

原文截图: Stable Diffusion 3 / MMDiT: 双流到单流的多模态 Transformer, Figure 2, MMDiT architecture; 来源 PDF 第 5 页。

Scaling Rectified Flow Transformers for High-Resolution Image Synthesis

Patrick Esser* Sumith Kulal Andreas Blattmann Rahim Entezari Jonas Müller Harry Saini Yam Levi
Dominik Lorenz Axel Sauer Frederic Boesel Dustin Podell Tim Dockhorn Zion English
Kyle Lacey Alex Goodwin Yannik Marek Robin Rombach*
Stability AI

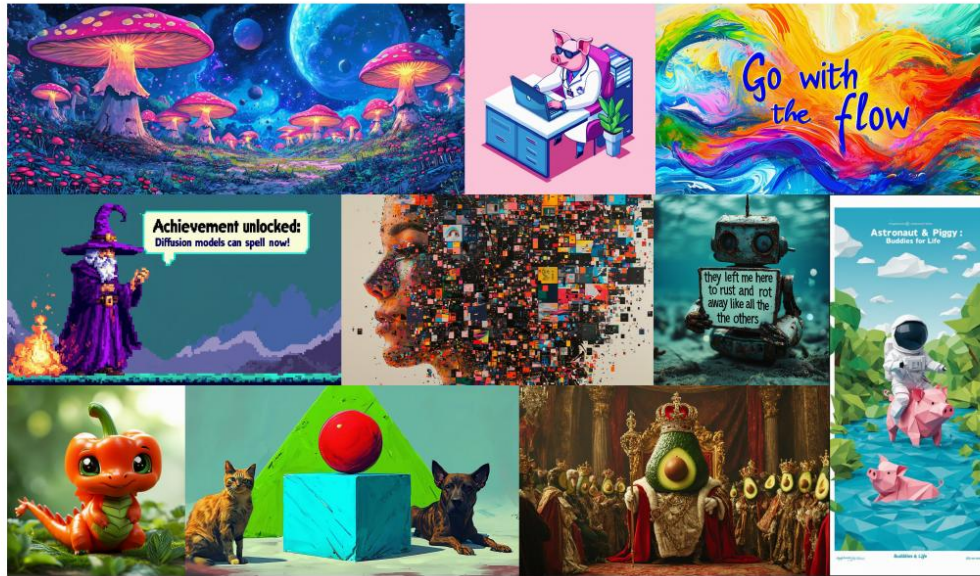


Figure 1. High-resolution samples from our 8B rectified flow model, showcasing its capabilities in typography, precise prompt following and spatial reasoning, attention to fine details, and high image quality across a wide variety of styles.

Abstract

Diffusion models create data from noise by inverting the forward paths of data towards noise and have emerged as a powerful generative modeling technique for high-dimensional, perceptual data such as images and videos. Rectified flow is a recent generative model formulation that connects data and noise in a straight line. Despite its better theoretical properties and conceptual simplicity, it is not yet decisively established as standard practice. In this work, we improve existing noise sampling techniques for training rectified flow models by biasing them towards perceptually relevant scales. Through a large-scale study, we demon-

strate the superior performance of this approach compared to established diffusion formulations for high-resolution text-to-image synthesis. Additionally, we present a novel transformer-based architecture for text-to-image generation that uses separate weights for the two modalities and enables a bidirectional flow of information between image and text tokens, improving text comprehension, typography, and human preference ratings. We demonstrate that this architecture follows predictable scaling trends and correlates lower validation loss to improved text-to-image synthesis as measured by various metrics and human evaluations. Our largest models outperform state-of-the-art models, and we will make our experimental data, code, and model weights publicly available.

*Equal contribution. <first.last>@stability.ai.

原文截图: Stable Diffusion 3 / MMDiT: 双流到单流的多模态 Transformer, Figure 1, high-resolution samples; 来源 PDF 第 1 页。

创新点与真正解决的问题 | Contribution

把 ‘模态特定表达能力’ 与 ‘跨模态交互’ 拆开，再在注意力层共享信息。FLUX.1 的 dual-stream / single-stream 组织可用这一脉络理解，但具体层数与参数不能由 SD3 推断。

它真正解决的问题是：提升文本遵循、文字生成和高分辨率扩散 Transformer 的训练稳定性。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

SD3 报告不能作为 FLUX.1 未公开参数的替代来源；二者应区分为相邻技术路线。

双流设计启示相机复原：图像条件应保持独立的空间表征，文本/任务提示作为语义控制端，融合点决定结构保真与风格生成的平衡。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

FLUX.1 Kontext : 以 Flow Matching 实现上下文图像编辑

论文角色	FLUX.1 Kontext 官方技术信息
作者/机构	Black Forest Labs; S. Batifol et al. / Black Forest Labs
发表状态	Technical report; arXiv:2506.15742
纳入等级	FLUX 结构核心

研究问题与前序缺陷 | Research question

传统 text-to-image 编辑往往需要重建整幅图像，导致身份漂移、局部编辑不稳定和多轮编辑累积误差。

方法拆解 | Method

将输入图像与 context image 编码为 latent tokens，与 text tokens 一起进入 FLUX 的多模态 Transformer；上下文图像参与 flow trajectory 的条件化，支持 in-context editing、style reference 和 iterative editing。

关键公式 | Key equation: $z_t = (1-t)z_0 + t z_1$; $v_\theta(z_t, t, c_{\text{text}}, c_{\text{image}}) \approx dz_t/dt$

Model	PDist ↓	SSIM ↑	PSNR ↑
Flux-VAE	0.332 ± 0.003	0.896 ± 0.004	31.1 ± 0.08
SD3-VAE [12]	0.452 ± 0.004	0.858 ± 0.005	29.6 ± 0.07
SD3-TAE ⁴	0.746 ± 0.004	0.774 ± 0.014	27.9 ± 0.06
SDXL-VAE [40]	0.890 ± 0.005	0.748 ± 0.006	25.9 ± 0.07
SD-VAE ⁵	0.949 ± 0.005	0.720 ± 0.004	25.0 ± 0.07

Table 1: Reconstruction quality comparison across different VAE architectures. All metrics computed on 4096 ImageNet images. Values are mean ± standard error (rounded). See also Appendix B.

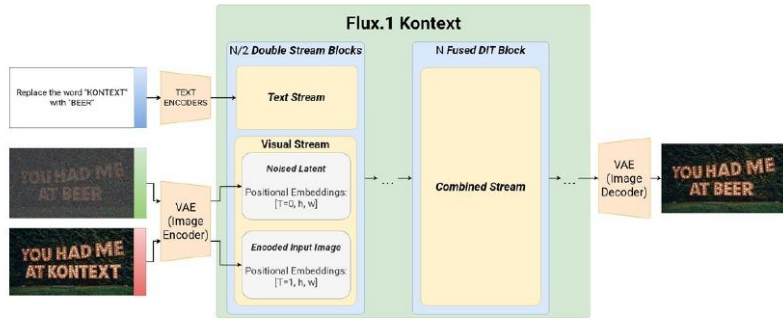


Figure 4: High-level overview of *FLUX.1 Kontext*, with input and context image on the left. Details in Section 3.

To that end, let $x \in \mathcal{X}$ be an output (target) image, $y \in \mathcal{X} \cup \{\emptyset\}$ an optional *context* image, and $c \in \mathcal{C}$ a text prompt. We model the conditional distribution $p_\theta(x | y, c)$ such that the same network handles *in-context* and *local edits* when $y \neq \emptyset$ and free *text-to-image generation* when $y = \emptyset$. Training starts from a FLUX.1 text-to-image checkpoint, and we collect and curate millions of relational pairs $(x | y, c)$ for optimization. In practice, we do not model images in pixel space but instead encode them into a token sequence as discussed in the following paragraph.

Token sequence construction. Images are encoded into latent tokens by the frozen FLUX auto-encoder. These context image tokens y are then appended to the image tokens x and fed into the visual stream of the model. This simple *sequence concatenation* (i) supports different input/output resolutions and aspect ratios, and (ii) readily extends to multiple images $y_1, y_2, \dots, y_N$. Channel-wise concatenation of x and y was also tested but in initial experiments we found this design choice to perform worse.

We encode positional information via 3D RoPE embeddings, where the embeddings for the context y receive a constant offset for all context tokens. We treat the offset as a *virtual time step* that cleanly separates the context and target blocks while leaving their internal spatial structure intact. Concretely, if a token position is denoted by the triplet $\mathbf{u} = (t, h, w)$, then we set $\mathbf{u}_x = (0, h, w)$ for the target tokens and for context tokens we set

$$\mathbf{u}_{y_i} = (i, h, w), \quad i = 1, \dots, N, \quad (2)$$

Rectified-flow objective. We train with a rectified flow-matching loss

$$\mathcal{L}_\theta = \mathbb{E}_{t \sim p(t), x, y, c} [\|v_\theta(z_t, t, y, c) - (\varepsilon - x)\|_2^2], \quad (3)$$

where z_t is the linearly interpolated latent between x and noise $\varepsilon \sim \mathcal{N}(0, 1)$; $z_t = (1-t)x + t\varepsilon$. We use a logit normal shift schedule (see Appendix A.2) for $p(t; \mu, \sigma = 1.0)$, where we change the mode μ depending on the resolution of the data during training. When sampling pure text-image pairs ($y = \emptyset$) we omit all tokens y , preserving the text-to-image generation capability of the model.

原文截图: FLUX.1 Kontext: 以 Flow Matching 实现上下文图像编辑, Figure 4, FLUX.1 Kontext overview; 来源 PDF 第 5 页。

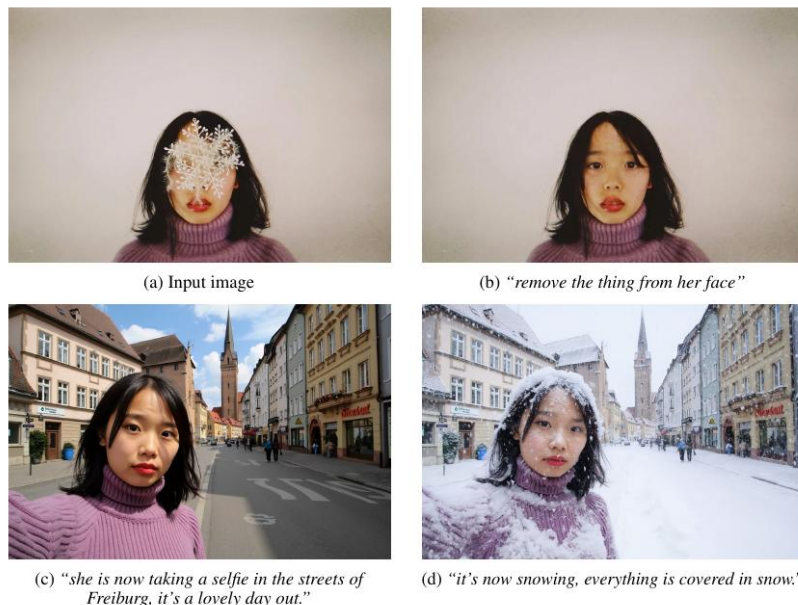


Figure 2: **Iterative, instruction-driven editing.** Starting from a reference photo (a), our model successively applies three natural-language edits—first removing an occlusion (b), then relocating the subject to Freiburg (c), and finally transforming the scene into snowy weather (d). Character, pose, clothing, and overall photographic style remain consistent throughout the sequence.

- **Generative editing.** Extraction of a visual concept (e.g. a particular figure or logo), followed by its faithful reproduction in new environments, potentially synthesized under a new viewpoint or rendering in a new visual context. Similarly to *in-context learning* in large language models, where the network learns a task from the examples provided in the prompt without any parameter updates [6], the generator adapts its output to the conditioning context on the fly. This property enables personalization of generative image and video models without the need for finetuning [43] or LoRA training [19, 27, 20]. Early works on such training-free subject-driven image synthesis include *IP-Adapter* [58] or retrieval-augmented diffusion variants [7, 3].

Recent Advances. InstructPix2Pix [5] and subsequent work [4] demonstrated the promise of synthetic instruction-response pairs for fine-tuning a diffusion model for image editing, while learning-free methods for personalized text-to-image synthesis [13, 43, 24] enable image modification with off-the-shelf, high-performance image generation models [42, 40]. Subsequent instruction-driven editors such as Emu Edit [51], OmniGen [56], HiDream-EI [15] and ICEdit [62] – extend these ideas to refined datasets and model architectures. Huang et al. [20] introduce in-context LoRAs for diffusion transformers on specific tasks, where each task needs to train dedicated LoRA weights. Novel proprietary systems embedded in multimodal LLMs (e.g., GPT-Image [36] and Gemini Native Image Gen [23]) further blur the line between dialog and editing. Generative platforms such as Midjourney [35] and RunwayML [44] integrate these advances into end-to-end creative workflows.

Shortcomings of recent approaches. In terms of results, current approaches struggle with three major shortcomings: (i) instruction-based methods trained on synthetic pairs inherit the shortcomings of their generation pipelines, limiting the variety and realism of achievable edits; (ii) maintaining the accurate appearance of characters and objects across multiple edits remains an open problem, hindering story-telling and brand-sensitive applications; (iii) in addition to lower quality compared to

原文截图: FLUX.1 Kontext: 以 Flow Matching 实现上下文图像编辑, Figure 2, iterative editing; 来源 PDF 第 3 页。

创新点与真正解决的问题 | Contribution

将图像上下文作为一等条件, 而非先生成描述再重新采样。对后续 SR/IR 论文而言, FLUX 不只是生成器, 也成为可通过 LoRA/ControlNet/条件 latent 适配的图像变换底座。

它真正解决的问题是：改善多轮编辑的一致性和局部控制能力。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

官方报告披露的工程细节有限；模型卡与论文中可见信息不能替代完整训练配方、数据许可或商业部署测试。

手机相册中的‘保持人物/主体不变，仅修复细节’可以借鉴 context conditioning；但相机应用必须额外约束纹理真实性和元数据可信度。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

FLUX 模型卡核验状态 | Model-card verification

模型卡属于版本、许可证、用途边界和已知限制的主要证据。本次尝试直接保存 Black Forest Labs 在 Hugging Face 的公开 README，但当前网络连接连续超时，因此下表只记录官方入口和核验状态，不转述未下载内容。

模型	官方入口	当前状态
FLUX.1-dev	https://huggingface.co/black-forest-labs/FLUX.1-dev	待联网下载并逐条核验
FLUX.1-schnell	https://huggingface.co/black-forest-labs/FLUX.1-schnell	待联网下载并逐条核验
FLUX.1-	https://huggingface.co/black-forest-labs/FLUX.1	论文证据已下载；模型卡

Kontext-dev	labs/FLUX.1-Kontext-dev	待核验
FLUX.2-dev	https://huggingface.co/black-forest-labs/FLUX.2-dev	FoA-SR 正文证明使用; 模型卡待核验

核验原则：在模型卡原文成功保存前，不对 commercial use、license compatibility、safety limitation、recommended hardware 或训练数据范围作确定陈述。

第二篇 FLUX 超分主线 | Super-resolution

四篇超分论文可以看成四个正交旋钮：FluxSR 主要压缩时间步，DreamSR 扩展空间感受野，OP4KSR 处理无分块 4K 的周期伪影，FoA-SR 处理恢复意图与质量偏好的冲突。它们并非在同一数据、同一指标、同一模型版本下比赛，结论必须按论文自身 protocol 阅读。

FluxSR : One Diffusion Step to Real-World Super-Resolution

论文角色	FLUX teacher + SR backbone
作者/机构	Jianze Li; Jiezhong Cao; Yong Guo; Wenbo Li; Yulun Zhang / Shanghai Jiao Tong University / Harvard University / South China University of Technology / Huawei Noah's Ark Lab
发表状态	arXiv:2502.01993
纳入等级	超分重点案例

研究问题与前序缺陷 | Research question

真实世界 SR 同时面临未知退化、纹理缺失和扩散模型推理步数高的问题。直接把 LR 图像送入通用 FLUX 会出现条件分布不匹配，且多步采样不适合实际部署。

方法拆解 | Method

FluxSR 使用预训练 FLUX 的多步 flow trajectory 作为教师，离线生成 noise-image pairs，并通过 Flow Trajectory Distillation (FTD) 训练一步 SR 模型；同时使用 reconstruction loss 与 adversarial/detail loss 约束重建。

关键公式 | Key equation: $v_{\theta}^L = (u_t - u_{\{t_L\}T_L}) / (1 - T_L)$; $L = L_{FTD} + \lambda_{rec} L_{rec} + \lambda_{ADL} L_{ADL}$

One Diffusion Step to Real-World Super-Resolution via Flow Trajectory Distillation

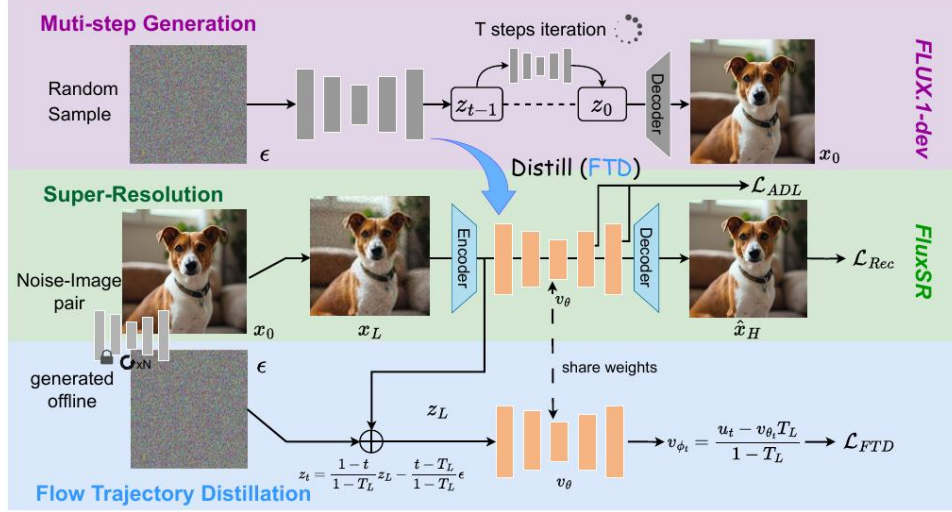


Figure 3: Training framework of FluxSR. (Top) Multi-step inference process of the pre-trained FLUX model. (Middle) Training strategy of FluxSR. (Bottom) Computation process of FTD. We distill a one-step super-resolution model from the multi-step FLUX model, without the need for the teacher model to be involved online during training.

4.2. Large Model Friendly Training Strategy

Although we have derived the theoretical formulation of FTD, its practical application faces the following challenges:

i) Inference Efficiency: During inference, we need both the vector field u_t calculated by the pre-trained T2I model and the vector field u_t^L calculated by the model fine-tuned on LR data. This requires two different flow models with separate parameters, leading to significant computational overhead during inference. **ii) Estimation Error:** Running the flow model in a one step makes it difficult to accurately estimate the velocity at time t . Without using a reconstruction loss to optimize the generator, the model performance may degrade. In this section, we propose an optimized training strategy to ensure that only a one flow model is required during inference. Additionally, we incorporate a reconstruction loss to enhance model performance.

Direct Parameterization of u_t^{SR} . As shown on the left side of Figure 3, since we can derive u_t^{SR} from u_t and u_t^L , we can also obtain u_t^L from u_t and u_t^{SR} . This avoids the issue caused by the inability to directly parameterize u_t^{SR} . We parameterize u_t^{SR} using v_θ . To represent both u_t^L and u_t^{SR} with a one model, we define the time step corresponding to the LR image as T_L instead of 0. This ensures that the model represents only u_t^L in the time range $[T_L, 1]$ and only u_t^{SR} at T_L . Additionally, the LR image distribution is more similar to the intermediate states x_t of the pre-trained diffusion

model. As shown in Figure 3, similar to Eq. (11), we have:

$$\begin{cases} x_0 = \epsilon - u_t, \\ x_L = \epsilon - (1 - T_L)u_t^L, \\ x_L - x_0 = u_t^{SR}T_L. \end{cases} \quad (13)$$

By combining the above equations, we obtain:

$$u_t^L = \frac{u_t - u_t^{SR}T_L}{1 - T_L}, \quad \text{where } t \in [T_L, 1]. \quad (14)$$

The model parameterization can be expressed as:

$$v_{\phi_t}(x_t, t) = \frac{u_t(x_t|\epsilon) - v_{\theta_t}(x_t, t)T_L}{1 - T_L}, \quad (15)$$

where

$$x_t = \frac{1-t}{1-T_L}x_L + \frac{t-T_L}{1-T_L}\epsilon, \quad t \in [T_L, 1]. \quad (16)$$

Generating noise-to-image flow for distillation. We pre-compute noise-sample pairs generated by FLUX and use them as training data, without relying on any real images. This approach offers two crucial benefits for large model training. 1) By using data pairs generated by the teacher model, we can directly compute $u_t(x_t|\epsilon) = \epsilon - x_0$, thus avoiding the estimation error during single-step inference. 2) The teacher model is not required for online inference during training, which significantly reduces GPU usage and

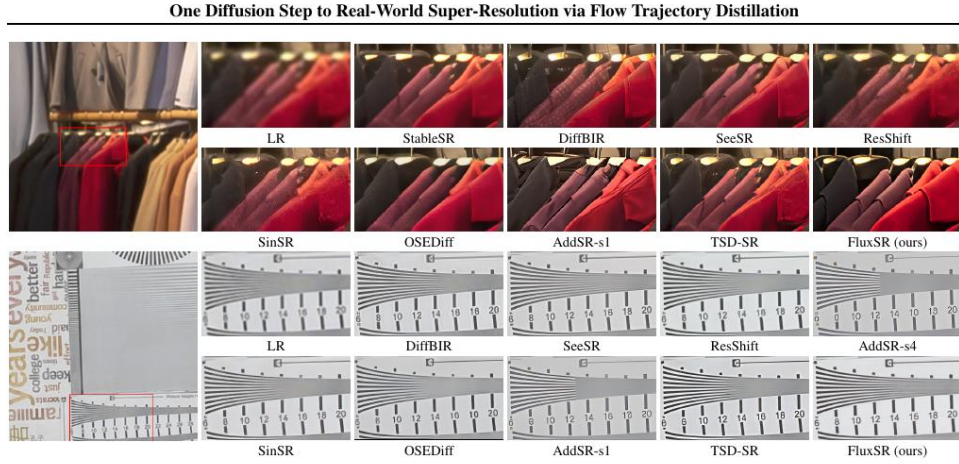
Figure 5: Visual comparisons ($\times 4$) on Real-ISR task.

Table 3: Ablation study on FTD.

Method	PSNR $\uparrow$	MUSIQ $\uparrow$	MANIQA $\uparrow$	Q-Align $\uparrow$
w/o FTD	26.33	56.02	0.3775	3.5170
FTD (ours)	24.67	67.84	0.5203	4.1473

Table 4: Ablation study on different loss functions.

$\mathcal{L}_{LPIPS}$	$\mathcal{L}_{TV-LPIPS}$	$\mathcal{L}_{EA-DISTS}$	$\mathcal{L}_{ADL}$	PSNR $\uparrow$	MUSIQ $\uparrow$	MANIQA $\uparrow$	Q-Align $\uparrow$
✓				23.10	64.55	0.4937	4.0515
	✓			22.09	65.04	0.5113	4.0927
		✓		23.67	64.83	0.5036	4.0003
			✓	24.72	67.13	0.5138	4.0691
	✓		✓	24.67	67.84	0.5203	4.1473

For example, in the first row of Figure 5, which depicts the restoration of a coat image, DiffBIR, ResShift, and SinSR are affected by noise, resulting in artificial textures. Although AddSR and TSD-SR generate relatively sharp images, they fail to accurately restore the collar's design. In contrast, FluxSR reconstructs the collar in a way that closely resembles the real-world appearance. The second row of Figure 5 demonstrates the restoration of numerical digits. FluxSR produces the most realistic result. While TSD-SR also approximately restores the digits, it suffers from Sinc noise, generating bright edges around the numbers.

5.3. Ablation Study

In this section, we use RealSR as the test dataset. The training iterations are set to 30k. Other settings remain consistent with those mentioned in Sec. 5.1.

Effectiveness of FTD loss. To verify the effectiveness and of FTD, we compare it with training using only the reconstruction loss, as shown in Table 3. Training the one-step flow model with only the reconstruction loss results in

poor performance, failing to generate high-frequency details and exhibiting significant high-frequency artifacts. Using the proposed FTD loss does not disrupt the data distribution learned by the teacher model. It effectively restores high-frequency details and achieves a higher degree of realism.

Effectiveness of ADL and TV-LPIPS. To verify the effectiveness of ADL and the proposed TV-LPIPS loss, we conducted relevant ablation experiments to investigate the impact of each loss function component. We also included the use of EA-DISTS, proposed by DFOSD, as a perceptual loss. Table 4 presents the experimental results, showing that using TV-LPIPS as a perceptual loss and ADL as a regularization term achieves the best performance.

6. Conclusion and Limitation

This paper proposes FluxSR, an efficient one-step Real-ISR model based on FLUX, the state-of-the-art T2I diffusion model. FluxSR leverages Flow Trajectory Distillation (FTD) to distill a multi-step flow matching model into a one-step super-resolution model. It is trained using noise-image pairs generated by a fixed multi-step model and does not require any real data. We employ TV-LPIPS and ADL to enhance high-frequency components in the generated images and reduce periodic artifacts. Our experiments demonstrate that FluxSR achieves unprecedented realism.

Limitations. Although FluxSR achieves strong performance, it has a large number of parameters and high computational cost. Moreover, we have not entirely eliminated the periodic artifacts mentioned in Section 4.3. In the future, we plan to apply model pruning techniques to compress the model and develop more effective algorithms to prevent periodic artifacts, aiming to achieve a lightweight yet high-performance Real-ISR model.

原文截图: FluxSR: One Diffusion Step to Real-World Super-Resolution, Figure 5, Real-ISR comparison; 来源 PDF 第 8 页。

创新点与真正解决的问题 | Contribution

教师模型在线推理被移到离线数据生成阶段, 使部署端只需一次 flow update; 同时通过轨迹中间状态构造 LR 条件, 避免单步速度估计不稳定。

它真正解决的问题是：把 FLUX 的生成先验压缩成 real-world SR 的单步模型，降低推理成本并缓解真实退化。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

一步生成天然更容易在 fidelity 与 perceptual quality 之间偏向某一侧；生成型纹理可能不是原始相机信息的可验证恢复。

适合云端照片精修和数字变焦首版：低延迟、可批量部署；端侧还需量化、蒸馏和显存/带宽评估。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 生成质量/复原 profile。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

DreamSR：面向超高分辨率的 receptive-field enhanced DiT

论文角色	Flux.1-dev backbone
作者/机构	Qingji Dong; Hang Dong; Mingqin Chen; Rui Zhang; Yitong Wang / ByteDance Inc.
发表状态	arXiv:2605.15682
纳入等级	超分重点案例

研究问题与前序缺陷 | Research question

patch-wise 4K SR 会产生 local over-generation、patch seams 与全局语义不一致；单纯增加局部窗口无法让模型看到足够大的 receptive field。

方法拆解 | Method

DreamSR 采用 degradation removal one-step process + texture generation multi-step process;
冻结/部分复用 FLUX DiT, 加入 Restoration Acceleration LoRA 与 Patch Context aware MM-ControlNet, 将 global context 和 local patch detail 一起注入。

关键公式 | Key equation: $z_t = (1-t)z_{HR} + t\epsilon$; $L = L_{rec} + \lambda_{ctx}L_{ctx} + \lambda_{tex}L_{tex} + \lambda_{adv}L_{adv}$

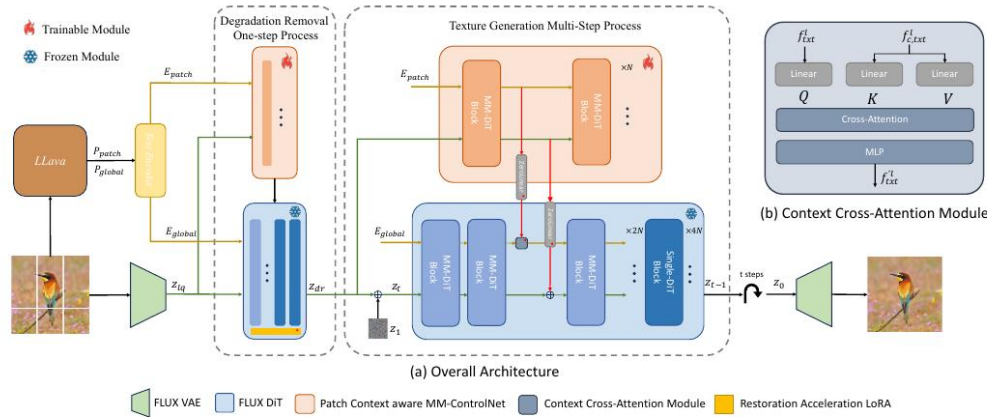


Figure 2. Overview of the proposed DreamSR architecture. Our framework consists of two stages: a degradation removal one-step process and a texture generation process. The Restoration Acceleration LoRA is introduced to compress the early denoising process, while the Patch Context aware MM-ControlNet integrates global and local textual features through the Context Cross-Attention module for effective semantic alignment and fine-grained detail generation.

Diffusion Models. Diffusion models [24] have demonstrated exceptional capability in generating high-quality and diverse natural images. Latent Diffusion Models (LDMs) [46] extend the diffusion process into latent space, greatly improving efficiency while maintaining fidelity in text-to-image synthesis. Through large-scale image-text pretraining, these models effectively capture complex image manifolds and semantic relationships. Several studies [10, 18, 45, 47, 63] have shown that text-to-image diffusion priors outperform GAN-based priors in modeling diverse natural images, motivating their adoption in high-quality super-resolution tasks. More recently, the Diffusion Transformer (DiT) architecture has emerged as a powerful alternative to conventional U-Net backbones, becoming the new mainstream design. In particular, multi-modal variants such as MM-DiT, used in models like FLUX [3] and SD3.0 [22], not only scale model capacity but also strengthen image-text interaction, resulting in more semantically aligned visual outputs. As a result, leveraging such powerful diffusion priors for high-fidelity and controllable image super-resolution has become an active research direction.

3. Methods

We build our DreamSR framework upon Flux.1-dev model [3], aiming to push the performance boundary of image super-resolution by leveraging the powerful pre-trained generative priors of a Diffusion Transformer (DiT). As shown in Figure 2, our DreamSR consists of a fixed DiT,

a Patch Context aware MM-ControlNet and a Restoration Acceleration LoRA [25] module. The reconstruction process is decomposed into two stages: degradation removal and texture generation. In the degradation removal stage, the Restoration Acceleration LoRA module is integrated into the DiT network, which compresses the early-stage denoising process and reduces inference computational cost. In the texture generation stage, the diffusion model (DiT & ControlNet) performs multi-step denoising to synthesize and refine high-frequency textures.

To mitigate the over-generation artifacts that arise from semantic mismatches between local patches and global text during patch-wise inference, we propose a Patch Context aware MM-ControlNet. Unlike previous methods [1] that inject only visual features from ControlNet into DiT, our approach integrates both visual and textual features from the MM-DiT blocks, where local textual information is seamlessly incorporated and effectively combined with global semantics. This design facilitates consistent semantic understanding across patches, ensuring coherent global structures and precise local texture reconstruction.

3.1. Patch Context aware MM-ControlNet

We construct our Patch Context aware MM-ControlNet $\mathcal{F}_\theta$ by partially copying the layers from the DiT network ϵ_ϕ in FLUX.1-dev [3]. To enhance the text interaction capability of $\mathcal{F}_\theta$, we selectively employ these text-related blocks (MM-DiT) while discarding Single DiT blocks. These text-related blocks can efficiently process patch-level textual in-

原文截图: DreamSR: 面向超高分辨率的 receptive-field enhanced DiT, Figure 2, DreamSR architecture; 来源 PDF 第 3 页。

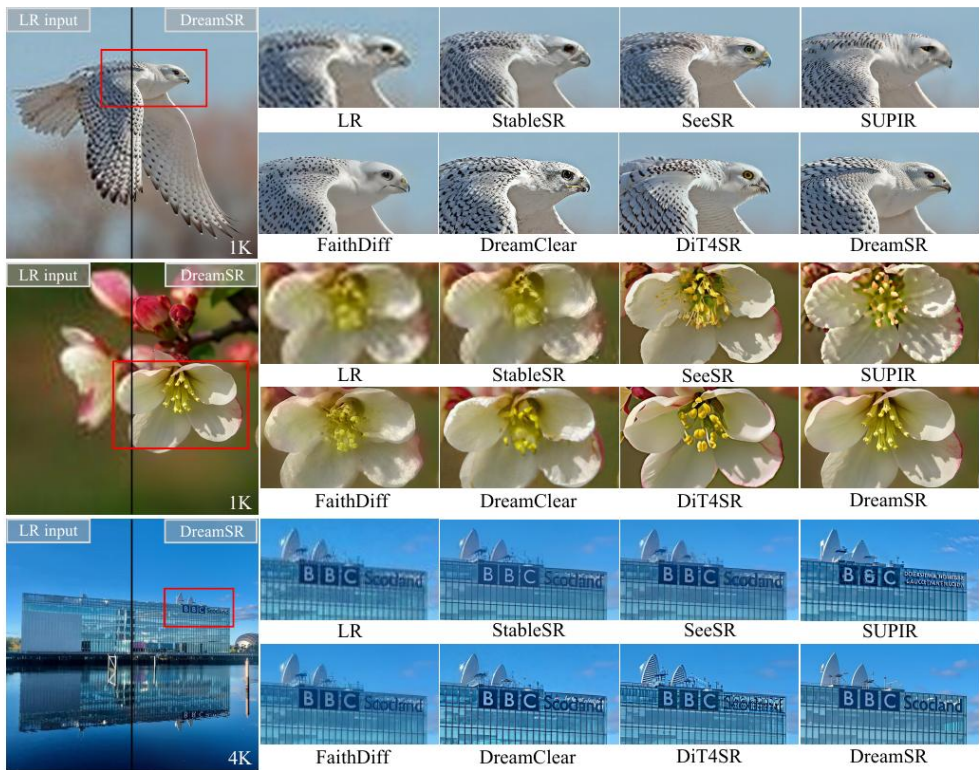


Figure 4. Qualitative comparisons with different methods on real-world datasets. Our DreamSR achieves the best performance, generating more realistic images with fine-scale structures and details. More visual results are included in the supplementary material.

Table 2. Inference-time performance of diffusion-based SR methods for the diffusion process on 2560x1440 resolution images. All methods are tested on a H20 GPU. The best and second performances are highlighted in red and blue, respectively.

	StableSR	DiffBIR	SeeSR	SUPIR	FaithDiff	DreamClear	DiT4SR	Ours
Step	200	50	50	50	20	50	40	1+16
Time (s)	156	127	119	124	30	185	228	86

are unable to produce fine-grained details, leading to overly smooth reconstructions. In contrast, our approach produces structurally coherent and detailed reconstructions, demonstrating its robustness and superior visual fidelity.

Inference time Comparisons. We evaluate the inference efficiency of our model on standard 2K images using a single H20 GPU. Benefiting from our acceleration design, the proposed method achieves high-quality detail reconstruction with only 1+16 inference steps. Our step count is significantly lower than that of other diffusion-based SR meth-

ods. Despite the relatively large parameter size of the DiT architecture, our model still runs faster than most methods built upon SD [46] or SDXL [43] UNet backbones, and is only slower than FaithDiff, which adopts a lightweight design. Overall, our approach achieves a favorable balance between inference speed and reconstruction quality.

4.3. Ablation Study

Effectiveness of dual-branch MM-ControlNet. To validate the effectiveness of the proposed dual-branch MM-ControlNet, we conduct an ablation study by replacing our MM-ControlNet with a conventional ControlNet architecture that removes the text feature branch. The model is trained under the same settings as the full model, except that it uses global prompt. During testing, we evaluate two variants: one using only the global prompt and the other using only patch-level prompts as both DiT and Control-

38264

原文截图: DreamSR: 面向超高分辨率的 receptive-field enhanced DiT, Figure 4, qualitative comparison; 来源 PDF 第 7 页。

创新点与真正解决的问题 | Contribution

把 ‘退化去除’ 与 ‘纹理生成’ 拆成不同计算阶段，并用 patch-context-aware cross-attention 连接局部块与全局语义，从而扩展到 ultra-high-resolution。

它真正解决的问题是：缓解 patch-based 推理的局部过生成和跨块不一致，同时压缩前期 denoising 成本。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

多阶段推理仍有成本；超高分辨率下的纹理真实性、极端退化和不同相机 ISP pipeline 的泛化需要独立验证。

适合云端 2K/4K 精修和长焦数字变焦的细节增强；对手机端，应优先迁移早期加速模块并固定输出 profile。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 生成质量/复原 profile。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

OP4KSR : One-Step Patch-Free 4K SR with Periodic Artifact Suppression

论文角色	Flux-based backbone
作者/机构	Chengyan Deng; Pengbin Yu; Zhentao Chen; Wei Shen; Kai Zhang; Meng Li; Lunxi Yuan; Xue Zhou; Li Yu / UESTC / OPPO AI Center, OPPO Inc. / Nanjing University
发表状态	arXiv:2605.13457
纳入等级	超分重点案例

研究问题与前序缺陷 | Research question

4K patch-free 推理中，RoPE 在超出训练分辨率后出现 phase collapse；一步 latent alignment 又会放大 periodic/grid artifacts。

方法拆解 | Method

在 UltraFlux/FLUX 体系上采用 compact F16 VAE 与 one-step flow matching; 通过 RoPE Base Frequency Rescaling (RFR) 延缓空间频率衰减, 并引入 Autocorrelation-based Periodicity Loss (L_{AP}) 抑制周期性伪影。

关键公式 | Key equation: $z_t = t z_{HR} + (1-t)\epsilon$; $L = L_{vae} + L_{MSE} + L_{LPIPS} + L_{wav} + L_{AP}$;
 $L_{AP} = ||AC(x) - AC(y)||$

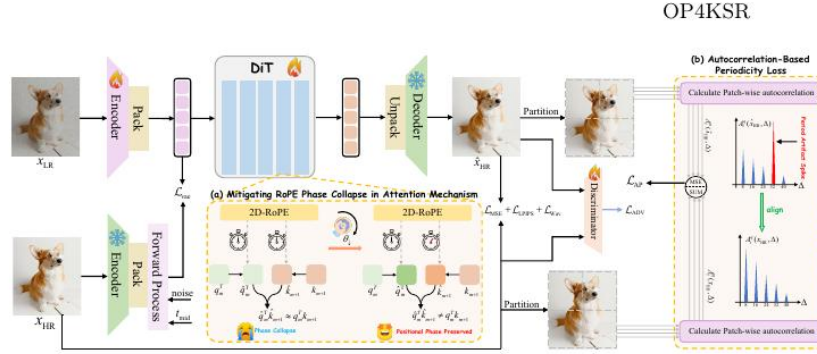


Fig. 4: Overall architecture of OP4KSR. To suppress periodic artifacts, we introduce two components: (a) RoPE Base Frequency Rescaling (RFR), which lowers the RoPE base frequency θ to resolve inter-token ambiguity and restore fine-grained structural sensitivity; (b) Autocorrelation-based Periodicity Loss (L_{AP}), which partitions images into $Q = 4$ quadrants for independent autocorrelation computation to penalize periodic patterns. Detailed formulations of L_{vae} [53], L_{MSE} , L_{LPIPS} , L_{wav} , and L_{ADV} are provided in the Supplementary Material.

real-world scenarios. Meanwhile, most real-world SR approaches [11, 45, 71, 77] are trained at low resolutions and still rely on region-wise inference when scaling to 4K, thereby inheriting the semantic and spatial limitations discussed in Section 1.

3 Method

3.1 One-Step 4K Super-Resolution Framework

To achieve patch-free, one-step 4K inference, OP4KSR builds upon UltraFlux [57], utilizing its highly compressed F16 VAE encoder E to bypass standard F8 constraints. Under the Flow Matching [33] framework, the intermediate latent z_t linearly interpolates between pure noise ϵ and the HR latent $z_{HR} = E(HR)$:

$$z_t = t z_{HR} + (1-t)\epsilon. \quad (1)$$

Adapting the pre-trained T2I framework for SR requires aligning the LR latent $z_{LR} = E(LR)$ to an appropriate point on this flow. Naively injecting z_{LR} at the initial timestep creates a severe distribution gap, as the model expects pure noise rather than structured image content, disrupting its learned generative prior [26]. To this end, we adopt the latent alignment strategy from OMGSR [53]. Specifically, we anchor z_{LR} at an optimal mid-timestep ($t_{mid} = 0.3$) to minimize the initial distribution gap, and further employ a latent alignment loss L_{vae} to explicitly constrain the intermediate latent z_t to match z_{LR} . This straightforward alignment seamlessly bridges the feature spaces, endowing OP4KSR with one-step SR capabilities. However, applying such one-step truncation to Flux-based 4K SR will introduce periodic artifacts. To resolve this, we introduce a periodic artifact suppression strategy: a tailored RoPE base frequency rescaling (RFR) as shown in Fig. 4(a) and an autocorrelation-based periodicity loss L_{AP} as shown in Fig. 4(b). The overall architecture of OP4KSR is detailed in Fig. 4.

原文截图: OP4KSR: One-Step Patch-Free 4K SR with Periodic Artifact Suppression, Figure 4, OP4KSR architecture; 来源 PDF 第 5 页。

OP4KSR 11

Table 3: Quantitative comparison on synthetic benchmarks (4KSR-Syn and DIV4K-50). The best and second-best results are highlighted in **bold** and underline.

Datasets	Metrics	Diffusion-based multi-step methods			Diffusion-based one-step methods				
		ResShift	SUPIR	DreamClear	SinSR	AddSR	OSDiff	OMGSR	OP4KSR
	T2I Prior	-	SDXL	PixArt- α	-	SD 2-base	SD 2.1-base	Flux.1	UltraFlux
4KSR-Syn	PSNR $\uparrow$	27.70	26.31	24.32	<u>27.06</u>	26.45	25.79	24.62	24.31
	SSIM $\uparrow$	0.7459	0.6884	0.6580	0.6990	<u>0.7342</u>	0.7200	0.6218	0.6830
	LPIPS $\downarrow$	0.4028	0.3571	0.3781	0.4208	0.3523	<u>0.3478</u>	0.4515	0.3297
	DISTS $\downarrow$	0.2055	<u>0.1533</u>	0.1617	0.1933	0.1858	0.1678	0.1935	0.1496
	TOPIQ_FR $\uparrow$	0.4311	0.4445	0.4059	0.4459	0.4007	<u>0.4472</u>	0.3958	0.4507
	NIQE $\downarrow$	5.82	<u>3.12</u>	2.84	5.57	4.78	3.77	5.38	4.80
	MANIQA $\uparrow$	0.3475	0.3840	<u>0.4103</u>	0.3734	0.4050	0.4587	0.3508	0.3931
	MUSIQ $\uparrow$	42.13	52.52	51.70	45.26	48.62	<u>55.10</u>	49.20	56.51
	CLIPQA $\uparrow$	0.5275	0.5776	0.5994	0.5925	0.5718	0.6578	0.5238	<u>0.6135</u>
	TOPIQ_NR $\uparrow$	0.4238	0.5034	0.5025	0.4694	0.5102	0.5803	0.4104	<u>0.5159</u>
	TOPIQ_IQA $\uparrow$	4.19	4.33	4.31	4.28	4.20	4.43	4.16	<u>4.37</u>
	NIMA $\uparrow$	4.41	4.51	4.51	4.48	4.28	<u>4.62</u>	4.52	4.68
DIV4K-50	PSNR $\uparrow$	26.21	24.73	23.33	<u>25.33</u>	24.85	24.72	23.80	23.09
	SSIM $\uparrow$	0.6304	0.5694	0.5554	0.5692	<u>0.6191</u>	0.6090	0.5292	0.5616
	LPIPS $\downarrow$	0.5506	<u>0.4640</u>	0.4744	0.5523	0.5193	0.4769	0.5389	0.4461
	DISTS $\downarrow$	0.2562	<u>0.1924</u>	0.2057	0.2324	0.2420	0.2161	0.2329	0.1843
	TOPIQ_FR $\uparrow$	0.3982	0.4131	0.3747	<u>0.4159</u>	0.3575	0.4096	0.3829	0.4263
	NIQE $\downarrow$	6.62	<u>3.14</u>	2.88	6.20	4.64	3.93	5.54	5.04
	MANIQA $\uparrow$	0.2643	0.3228	0.3648	0.3051	<u>0.3740</u>	0.4056	0.3077	0.3361
	MUSIQ $\uparrow$	33.36	46.81	47.46	38.03	47.71	<u>50.11</u>	42.68	52.24
	CLIPQA $\uparrow$	0.4345	0.5145	0.5379	0.5039	0.5267	0.6044	0.4630	<u>0.5409</u>
	TOPIQ_NR $\uparrow$	0.3464	0.4593	<u>0.5321</u>	0.4233	0.4987	0.5401	0.3724	0.4703
	TOPIQ_IQA $\uparrow$	4.11	4.44	4.32	4.27	<u>4.44</u>	4.57	4.14	4.29
	NIMA $\uparrow$	4.25	4.51	4.35	4.35	4.47	<u>4.48</u>	4.40	4.55

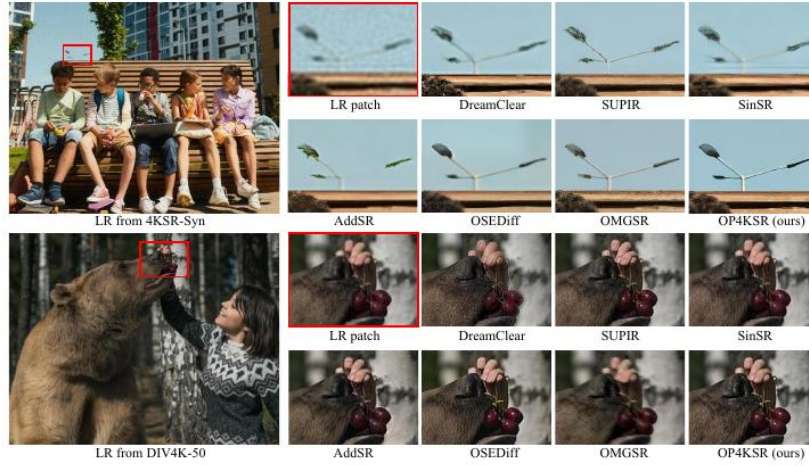


Fig. 8: Qualitative comparisons of different methods on two **synthetic datasets**.

and statistical metrics (e.g., PSNR and NIQE), deep-feature-based metrics (e.g., CLIPQA and LPIPS) are inherently resolution-sensitive. Thus, we adopt a patch-based evaluation strategy, as detailed in the Supplementary Material.

Implementation Details. Following [51, 53], we freeze the VAE decoder and exclusively fine-tune the VAE encoder and the DiT backbone using LoRA,

原文截图: OP4KSR: One-Step Patch-Free 4K SR with Periodic Artifact Suppression, Figure 8, qualitative comparison;

来源 PDF 第 11 页。

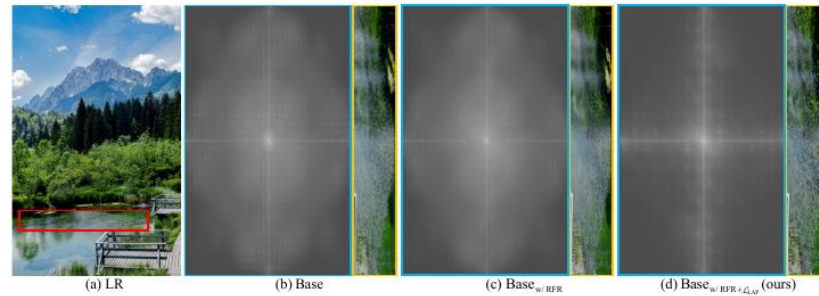


Fig. 10: Ablation study of the proposed artifact suppression mechanism in frequency and spatial domains. Blue boxes show the 2D FFT spectra of the full super-resolved images, while yellow boxes highlight local spatial details corresponding to the red box in (a). We compare: (a) the LR input, (b) baseline, (c) baseline + RFR, and (d) our full method (RFR + $\mathcal{L}_{AP}$).

4.3 Ablation

Ablation for Periodic Artifact Suppression Strategy. Table 6 verifies the effectiveness of our periodic artifact suppression strategy. Although the Base model achieves high PSNR, it suffers from severe periodic artifacts and limited perceptual quality. Introducing $\mathcal{L}_{AP}$ effectively suppresses grid anomalies, while RFR enhances structural fidelity by alleviating phase collapse. As they address complementary bottlenecks, their combination yields the best perceptual performance, confirming the necessity of both components.

To further validate the effectiveness of our proposed artifact suppression mechanism, we conduct a 2D FFT and spatial analysis in Fig. 10. As shown in Fig. 10(b), the baseline spectrum exhibits anomalous, grid-like high-frequency spikes, corresponding to the conspicuous woven-like periodic artifacts in the spatial domain. With RoPE base frequency rescaling (Fig. 10(c)), this abnormal energy concentration is partially diluted, yet the underlying periodic structures remain. Finally, synergizing RFR with our autocorrelation-based periodicity loss (Fig. 10(d)) successfully suppresses these anomalous spikes. The spectrum is restored to a smooth distribution, and spatial grid artifacts are suppressed.

5 Conclusion

This paper introduces OP4KSR, a patch-free one-step method for 4K super-resolution. Equipped with an F16 high-compression VAE, our method achieves exceptionally fast inference speeds and facilitates direct 4K outputs. To tackle the inherent periodic artifacts, we explicitly rectify the base frequency of 2D-RoPE and introduce a novel autocorrelation-based periodicity loss, resulting in substantial artifact suppression. Additionally, to propel the development of the 4K super-resolution task, we curate the 4KSR-Train dataset and establish three

原文截图: OP4KSR: One-Step Patch-Free 4K SR with Periodic Artifact Suppression, Figure 10, periodic-artifact ablation; 来源 PDF 第 14 页。

创新点与真正解决的问题 | Contribution

将 4K 周期伪影定位为 ‘位置编码频率’ 与 ‘patch 周期性统计’ 的共同问题，而不是简单归因于数据或 VAE；RFR + autocorrelation loss 形成结构与频域双约束。

它真正解决的问题是：在无分块、一步、4K 条件下同时处理空间感受野和周期性伪影。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

频率重标定可能改变不同宽高比的空间归纳偏置；论文中的真实相机退化覆盖和端侧 kernel 性能仍需更多证据。

最接近手机数字变焦的工程约束：patch-free 避免拼接边界，一步降低延迟；但必须保留文字、纹理和人脸等高风险区域的真实性检测。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 生成质量/复原 profile。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

FoA-SR：把 Faithful / Aesthetic 变成可选择的恢复 profile

论文角色	FLUX.2-dev backbone
作者/机构	Amjad Mahdi Alqarni; Peizhong Ju / University of Kentucky
发表状态	arXiv:2606.10275
纳入等级	超分重点案例

研究问题与前序缺陷 | Research question

真实 SR 中 PSNR/结构忠实与感知美学/纹理丰富经常冲突。单一模型输出无法同时满足文档、证件、风景和社交照片的不同偏好。

方法拆解 | Method

Stage 1 将 FLUX.2-dev 适配为共享 Flux2SR；Stage 2 用随机候选池构造 profile-specific preference pairs，分别训练 Faithful LoRA 和 Aesthetic LoRA，推理时选择一个 adapter。

关键公式 | Key equation: $L_{SR} = L_{FM} + \lambda_{pix} L_{charb} + \lambda_{perc} L_{LPIPS}$;

$\pi^F, \pi^A = \text{PreferenceOptimization}(\text{candidates}, R_F, R_A)$

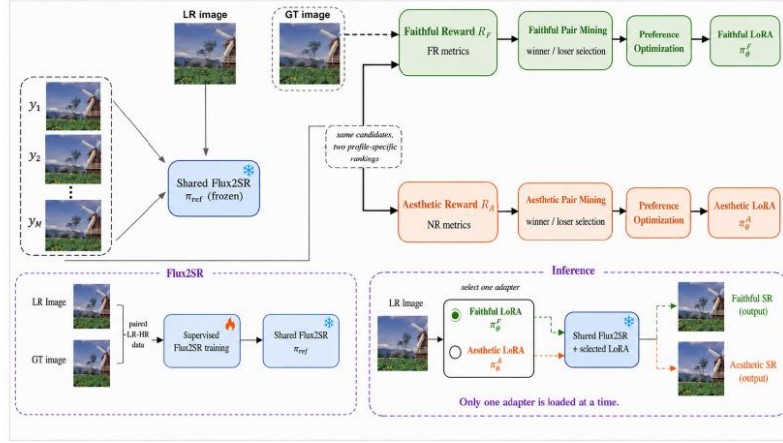


Figure 2: Overview of the proposed FoA-SR framework. Stage 1 adapts FLUX.2-dev into a shared supervised SR baseline, Flux2SR, using LR-HR paired supervision. Stage 2 uses the same Flux2SR model to generate a shared stochastic candidate pool for each LR input. The same candidates are ranked separately by Faithful and Aesthetic rewards to mine profile-specific winner-loser pairs. FoA-SR then fine-tunes separate LoRA adapters for the two profiles. At inference time, the user selects either the Faithful or Aesthetic adapter, exposing the restoration trade-off as a controllable choice rather than collapsing it into a single output.

realism may not preserve the reference exactly, while a model optimized for distortion metrics may produce conservative or less visually appealing outputs. FoA-SR operationalizes this trade-off as a profile-aware preference optimization problem. Instead of treating the trade-off as a single fixed balance, it exposes the restoration intent as an explicit choice between Faithful and Aesthetic adapters.

3 Method

FoA-SR consists of two stages. First, we adapt FLUX.2-dev into a paired real-world super-resolution model, which we call **Flux2SR**. Second, we use Flux2SR as a shared stochastic reference model to construct profile-specific preference pairs and fine-tune separate Faithful and Aesthetic adapters. The key difference from single-reward preference alignment is that FoA-SR does not collapse restoration quality into one universal objective; instead, it preserves Faithful and Aesthetic restoration as separate preference profiles. Figure 2 illustrates the overall FoA-SR pipeline.

3.1 Flux2SR: Adapting FLUX.2-dev for paired super-resolution

FLUX.2-dev is a large rectified-flow image generation backbone, but it is not directly a paired LR-to-HR super-resolution model [2, 3, 12]. We therefore build **Flux2SR**, a supervised SR adapter that conditions the FLUX.2 transformer on the low-resolution image while using the high-resolution image as the restoration target. Given an LR-HR pair (x, y^*) , we encode both images into the FLUX.2 latent space. The LR latent is packed as a conditioning image sequence, while the HR latent defines the flow-matching target.

We fine-tune only LoRA parameters in the FLUX.2 transformer [11] and keep the VAE, text encoder, and base transformer weights frozen. The supervised SR objective combines latent flow matching with image-space reconstruction losses:

$$\mathcal{L}_{SR} = \mathcal{L}_{FM} + \lambda_{pix} \mathcal{L}_{charb} + \lambda_{lpips} \mathcal{L}_{LPIPS}. \quad (1)$$

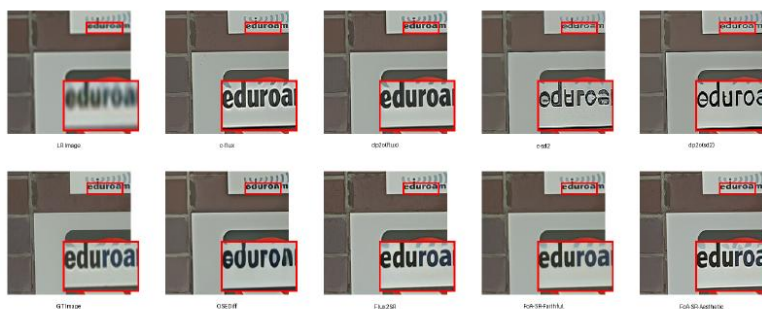


Figure 3: Qualitative comparison on a text-containing RealSR example. FoA-SR-Faithful emphasizes reference-consistent restoration, while FoA-SR-Aesthetic emphasizes sharper and more visually enhanced details.

Table 2: Hybrid reward ablation on RealSR. FoA-SR-Hybrid is trained with the averaged reward $R_H = 0.5R_F + 0.5R_A$. It improves several no-reference metrics but does not match the reference-consistency gains of FoA-SR-Faithful, supporting profile-specific optimization.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	DISTS $\downarrow$	CLIPQA $\uparrow$	NIQE $\downarrow$	MUSIQ $\uparrow$	MANIQA $\uparrow$	FID $\downarrow$
Flux2SR	22.7195	0.6733	0.2793	0.2191	0.6154	4.9360	69.5302	0.6667	110.0081
FoA-SR-Faithful	24.8633	0.7327	0.2502	0.2046	0.5535	5.1734	66.6408	0.6252	107.1397
FoA-SR-Aesthetic	20.9118	0.5695	0.4038	0.2743	0.7170	5.0522	73.0874	0.7270	123.7668
FoA-SR-Hybrid	22.6407	0.6539	0.3252	0.2522	0.7120	4.5909	72.6504	0.7026	121.1101

optimize a single universal quality score; instead, it steers the same Flux2SR baseline toward distinct restoration profiles.

Compared with single-output Real-ISR baselines such as DP²O-SR and OSediff, FoA-SR provides a complementary capability: it exposes restoration intent as an explicit choice between Faithful and Aesthetic adapters rather than committing to one fixed balance between fidelity and perceptual enhancement.

Qualitative comparison. Figure 3 shows a qualitative comparison on a text-containing RealSR example. FoA-SR-Faithful preserves the text structure more conservatively and avoids strong distortions, while FoA-SR-Aesthetic produces a sharper and more visually enhanced appearance but may introduce stronger local changes. Additional qualitative comparisons on RealSR and DIV2K-Val are shown in Appendix D.

4.3 Analysis and Ablations

Hybrid reward ablation. To test whether Faithful and Aesthetic restoration can be replaced by a single scalar objective, we train an additional FoA-SR-Hybrid adapter using the same Flux2SR initialization, candidate pool, and preference optimization setup. The only difference is that winner-loser pairs are mined using a hybrid reward:

$$R_H(y_i) = 0.5R_F(y_i) + 0.5R_A(y_i).$$

As shown in Table 2, FoA-SR-Hybrid improves several no-reference metrics, including CLIPQA, NIQE, MUSIQ, and MANIQA, but it does not recover the strong reference-consistency gains achieved by FoA-SR-Faithful. This supports our central design choice: averaging Faithful and Aesthetic rewards into a single scalar objective produces an implicit compromise, whereas FoA-SR exposes the two endpoints as selectable restoration profiles. Extended held-out metrics for the Hybrid adapter are reported in Appendix C.3.

原文截图：FoA-SR：把 Faithful / Aesthetic 变成可选择的恢复 profile，Figure 3, faithful/aesthetic comparison；来源 PDF 第

创新点与真正解决的问题 | Contribution

不把 fidelity-aesthetic trade-off 压成一个权重，而是显式建模成两个可选 restoration profiles。

它真正解决的问题是：让用户或业务策略在忠实恢复与审美增强之间做可解释选择。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

候选池和 reward metric 会决定偏好结果；Aesthetic profile 更可能引入不可验证纹理，不宜默认用于证件、医疗或取证。

手机相机可按场景提供 ‘真实’ 与 ‘增强’ 两个模式；相册/云端应保存 profile、模型版本和是否生成纹理等审计信息。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 生成质量/复原 profile。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

超分路线横向比较 | Cross-paper comparison

论文	FLUX 角色	主要计算策略	空间策略	质量控制	更适合的业务
FluxSR	FLUX teacher/bac kbone	FTD, one-step	常规 SR latent	Rec + perceptual	低延迟云端精修
DreamSR	Flux.1-dev	早期 LoRA 加速 + 多步纹理	patch context	global/local context	2K/4K 高质精修
OP4KSR	Flux-based	one-step	patch-free 4K	RFR + L_AP	无拼接边界数字变焦
FoA-SR	FLUX.2-dev	LoRA + preference	shared Flux2SR	Faithful/Aesthetic	可选恢复 profile

第三篇 FLUX 图像复原扩展 | Image restoration

复原篇的共同问题是：生成模型具有强先验，但真实退化的逆映射需要任务条件。FluxIR 通过 acquire-then-adapt 解决数据与适配，Edit2Restore 通过 few-shot LoRA 解决新任务快速迁移，LucidFlux 通过 caption-free 设计减轻语言条件污染，ResFlow-Tuner 则把推理时计算预算用于不同难度的输入。

Acquire and then Adapt / FluxIR：先从 FLUX 获取先验，再做轻量适配

论文角色	FLUX data generator + FluxIR backbone
作者/机构	Junyuan Deng; Xinyi Wu; Yongxing Yang; Congchao Zhu; Song Wang; Zhenyao Wu / Honor Device Co., Ltd / Shenzhen University of Advanced Technology
发表状态	arXiv:2504.15159
纳入等级	复原重点案例

研究问题与前序缺陷 | Research question

真实复原缺少高质量 paired data，而直接把大模型当 restoration model 又会出现任务条件不匹配。

方法拆解 | Method

FluxGen 用空 prompt 的 FLUX 生成高质量 clean images，构造 synthetic degradation pairs；FluxIR 采用单个/少量 MM-DiT block、可学习 T5 embedding 与 SE layers 进行轻量适配。

关键公式 | Key equation: $L=L_{FM}+\lambda_{rec}L_{rec}+\lambda_{LP}L_{LP}$; $z_t=(1-t)z_{clean}+t\epsilon$

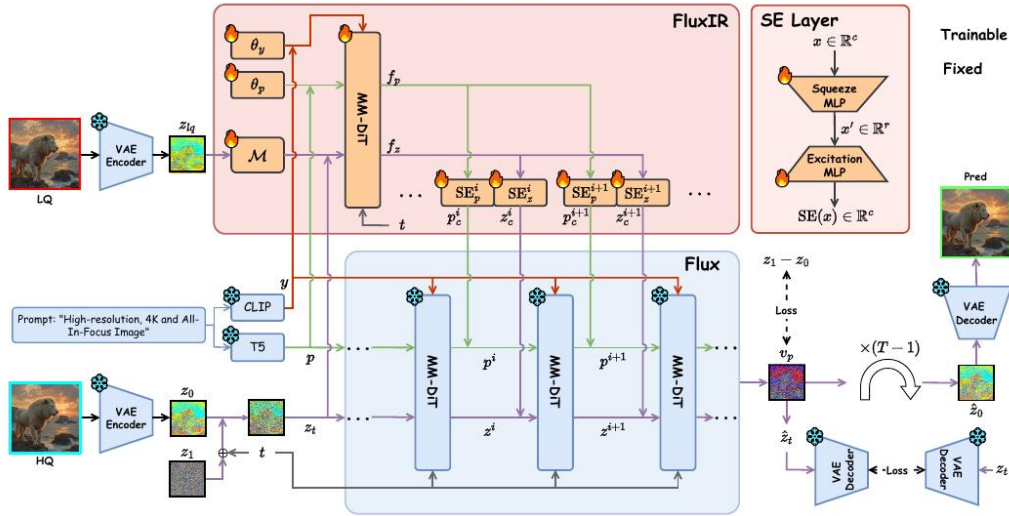


Figure 3. Training and inference pipeline of the proposed FluxIR. FluxIR employs a single MM-DiT block with learnable T5 embedding θ_p and CLIP embedding θ_y to extract image feature f_z and text feature f_p from the low-quality control latent z_{lq} . The squeeze-and-excitation (SE) layers ($SE_z(\cdot)$ for image and $SE_p(\cdot)$ for text) broadcast these features to all Flux MM-DiT blocks to enable precise and multi-modality control.

animals, plants, and the moon) or particular domains (e.g., nighttime scenes and aerial photography).

Image Selection. Although Flux is a powerful text-to-image model, the stability of the images it generates is not guaranteed. Low-quality images can significantly affect the training of image restoration models. To address this, we distill the dataset using various non-reference image quality assessment (IQA) models, including CLIP-IQA [74], MANIQA [91], and MUSIQ [32]. Specifically, we select images with CLIP-IQA, MANIQA, and MUSIQ scores within the top 95% as our training set. This process ensures that low-quality and failed generated images are removed from the training dataset, thus significantly enhancing our image restoration performance.

Pair-data Construction. The degradation of ground truth images has been widely studied in [77, 102], with common degradation types including blur, downsampling, noise, JPEG compression and *etc.* In this paper, we apply the synthetic degradation method [77] with the same settings as [85] to construct paired data for training.

3.2. FluxIR Architecture

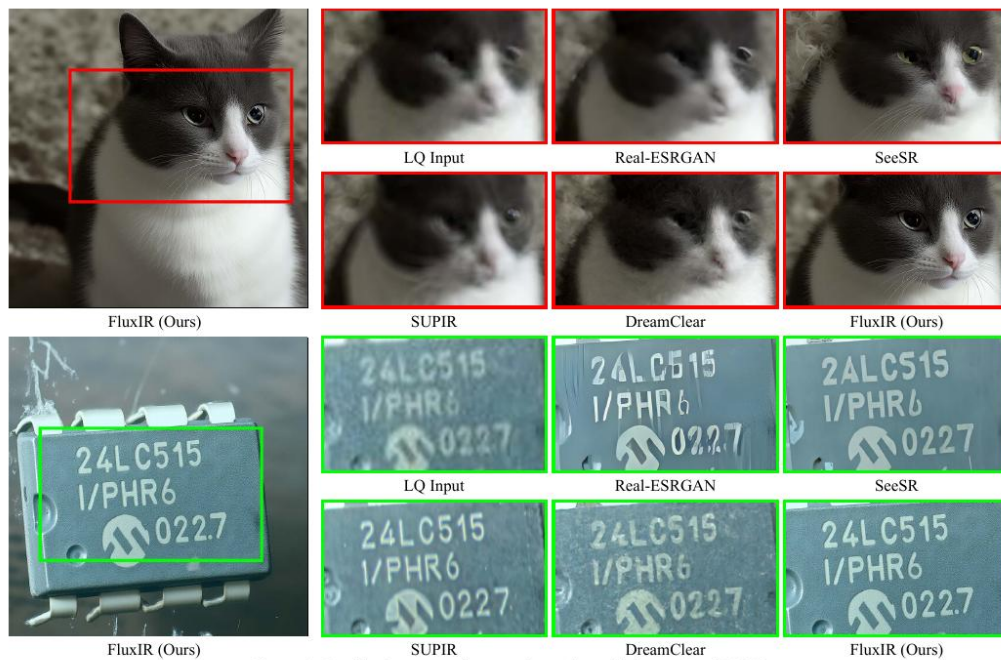
A typical solution for the T2I-based image restoration is training a ControlNet [103] upon a T2I model, with a VAE encoder to project the input image into latent space. The ControlNet is initialized as a copy of the U-Net encoder,

with zero convolution layers acting as bridges to integrate conditional controls. Then the extracted multi-level features are injected into corresponding layers of the U-Net decoder. Recently, DreamClear [3] introduced a DiT-based ControlNet model by duplicating all 28 DiT blocks from PixArt- α [10]. With 2.2B trainable parameters in ControlNet, it requires 224 GPU days for training, which is surprisingly time-consuming. To solve this issue, we carefully design a training-friendly 0.4B adapter, integrated with Flux — a much larger MM-DiT-based text-to-image model with approximately 12B parameters.

The detailed FluxIR architecture is illustrated in Fig. 3. To extract control signal from VAE embedding feature z_{lq} , only one MM-DiT block is involved in our FluxIR adapter for lightweight purposes. The MM-DiT block, denoted as $\mathcal{D}(z, p, y, t; \Theta_d)$, is initialized from the first MM-DiT block of the pre-trained Flux model, and its input can be reformulated as

$$z_c = z_t + \mathcal{M}_i(z_{lq}; \Theta_i), \quad (1)$$

where $\mathcal{M}_i(\cdot; \Theta_i)$ is a multi-layer perceptron (MLP) with parameters Θ_i initialized to zero and $z_t = (1 - t)z_0 + tz_1$ represents the linear combination of the noise latent z_1 and the ground truth latent z_0 at timestep t . To bridge the gap between the adapter and original T2I model, we employ a learnable T5 [58] embedding θ_p and a learnable CLIP [57]

Figure 5. Qualitative comparison on the real-world dataset *RealLQ250*.

RealESRGAN[77], SwinIR [42] and DASR[43]) and recent generative IR models (StableSR [75], DiffBIR [46], ResShift [95], SinSR [78], SeeSR [85], SUPIR [94], DreamClear [3]).

Quantitative Comparisons. Tab. 1 shows the quantitative comparisons with the existing SOTAs on the four datasets. Our FluxIR model achieves the best score in MANIQA with a significant margin across all synthetic and real-world datasets. On the *RealLQ250* dataset, our method significantly outperforms other methods across all non-reference metrics, while MANIQA even surpasses the second-best method by over 6.53%. On the other three synthetic and real-world datasets, our approach achieves the best or second-best scores in all non-reference metrics (CLPIQA, MUSIQ, and MANIQA). It is worth mentioning that our proposed method trained within 14 GPU-days significantly reduces training costs by approximately 93.5% and 91.5% compared to SUPIR [94] and DreamClear [3], respectively.

Qualitative Comparisons. We present a qualitative comparison on the synthetic dataset *DIV2K-Val* in Fig. 4. Our method demonstrates a notable advantage in generating high-frequency details, such as lip texture, hair, eyelashes, and pupils in portrait images. In contrast, SUPIR [94] and

DreamClear [3] fail to accurately reproduce human skin textures, and SeeSR [85] produces overly smooth results, lacking the natural skin texture and fine details. For the real-world dataset *RealLQ250*, we provide a qualitative comparison in Fig. 5. In the first row, which shows restoration results for a low-quality image of a cat, our method effectively reconstructs realistic high-frequency details such as fur, whiskers, and eyes. SeeSR [85] restores some fur and whiskers but lacks overall realism, while other methods fail to produce a high-quality, realistic result. In the second row, focusing on text recovery, our method achieves the most accurate text reconstruction, whereas other methods introduce various text distortions. Specifically, SUPIR [94] produces blurry results, and SeeSR [85] mistakenly recovers incorrect text content.

4.3. Ablation Study

Effectiveness of SE layer. We evaluate the impact of different ranks of our SE layer by setting r to 16, 32, 64, and 128, respectively. Another variant is replacing the SE layer with a single MLP, *i.e.* a full rank setting. As shown in Tab. 2, the SE layer with $r = 32$ achieves the best performance on CLPIQA, MANIQA, and CLPIQA. Compared to using full-rank MLPs, the SE layer with low-rank results in a comparable performance while saving over 72% of

原文截图: Acquire and then Adapt / FluxIR: 先从 FLUX 获取先验, 再做轻量适配, Figure 5, RealLQ250 comparison; 来源 PDF 第 7 页。

创新点与真正解决的问题 | Contribution

将大模型的价值拆为 data acquisition 与 task adaptation 两个阶段: 先利用 FLUX prior 生成训练数据, 再用低参数模块学习退化逆映射。

它真正解决的问题是: 缓解 paired restoration data 不足和全量微调成本高的问题。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

合成数据的退化分布可能与真实相机 ISP 不一致；空 prompt 生成内容的分布偏差会影响极端场景。

适合构建手机相机内部数据闭环：用真实 ISP 退化模拟器生成 paired data，再只更新 adapter，而不是重新训练大模型。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

Edit2Restore：用 few-shot LoRA 把编辑模型改造成复原模型

论文角色	FLUX.1 Kontext pretrained editing model
作者/机构	M. Akin Yilmaz; Ahmet Bilican; Burak Can Biner; A. Murat Tekalp / Codeway AI / Koc University
发表状态	arXiv:2601.03391
纳入等级	复原重点案例

研究问题与前序缺陷 | Research question

为每种噪声、雨、雾训练专用 restoration network 需要大量 paired data；通用编辑模型的零样本复原又不够精确。

方法拆解 | Method

在 FLUX.1 Kontext 上训练 LoRA adapters, 用简单指令 remove the noise/rain/haze from the image, 单任务仅需 16-128 对图像; 冻结大部分基座和 text encoders, 推理采用 28 integration steps.

关键公式 | Key equation: $W=W_0+\alpha BA$; $L_{FM}=E||v_{\theta}(z_t,t,c)-u_t||^2$

Edit2Restore 5

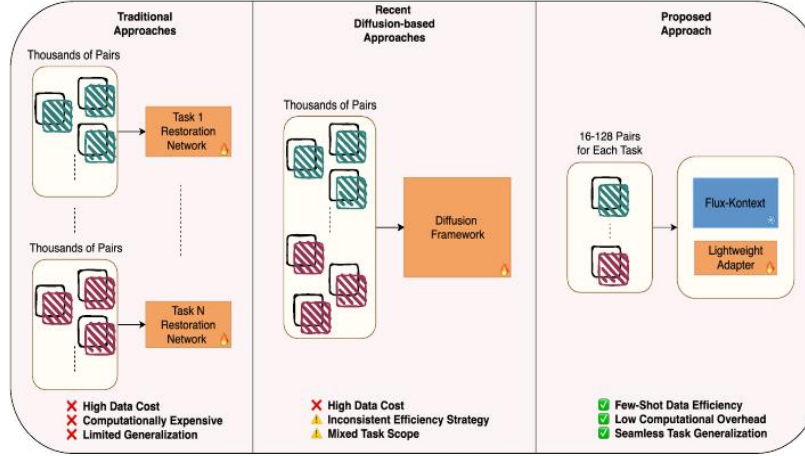


Fig. 1. Traditional approaches require training specific models with thousands of paired examples for each restoration task. Recent diffusion-based multi-task image restoration models leverage strong priors but still demand extensive paired data to train from scratch. Our approach achieves few-shot efficiency by adapting pre-trained image editing models with only 16-128 examples per task through parameter-efficient fine-tuning.

3.1 Preliminaries: FLUX.1 Kontext

FLUX.1 Kontext is a rectified flow transformer that performs image-to-image translation conditioned on both text instructions and input images. Given a degraded image $\mathbf{x}_{\text{deg}} \in \mathbb{R}^{H \times W \times 3}$ and a text prompt $\mathbf{t}$ describing the desired restoration operation, the model generates a restored image $\mathbf{x}_{\text{res}}$ through an iterative flow matching process in latent space.

The model consists of three main components that work together to generate images. A variational autoencoder with 16 latent channels encodes images into latent representations $\mathbf{z} = \mathcal{E}(\mathbf{x})$ and decodes them back to pixel space $\mathbf{x} = \mathcal{D}(\mathbf{z})$. Two text encoders, CLIP [36] and T5 [37], process the text prompt $\mathbf{t}$ into embeddings that condition the generation process. Finally, a flow matching transformer with a hybrid architecture combining double-stream and single-stream transformer blocks operates on the latent space, performing iterative refinement guided by text conditioning.

Unlike traditional diffusion models that predict noise, FLUX.1 Kontext uses rectified flow matching [25, 26] which learns to predict velocity fields. The forward process constructs a linear interpolation path between clean latent $\mathbf{z}_{\text{clean}}$ and Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$:

$$\mathbf{z}_t = (1 - t)\mathbf{z}_{\text{clean}} + t\epsilon \quad (1)$$

where $t \in [0, 1]$ is the timestep. The model learns to predict the velocity $\mathbf{v} = \epsilon - \mathbf{z}_{\text{clean}}$ that connects the two endpoints. This formulation provides straighter

原文截图: Edit2Restore: 用 few-shot LoRA 把编辑模型改造成复原模型, Figure 1, few-shot adaptation setting; 来源 PDF 第 5 页。

Edit2Restore 9

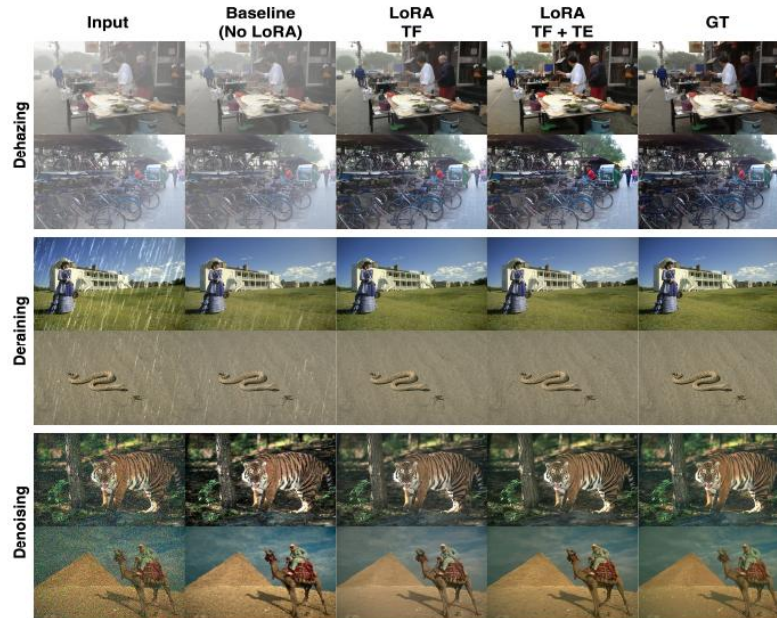


Fig. 2. Qualitative comparison of image restoration results.

purpose text understanding, while fine-tuning allows adaptation to restoration-specific semantics—potentially improving alignment between instructions and the corresponding visual transformations. Our results demonstrate consistent advantages from joint fine-tuning (LoRA: TF + TE). For dehazing, both FID and CMMD improve, indicating better distribution matching with ground truth images. Deraining shows similar trends with FID dropping from 12.36 to 11.76 and CMMD from 0.42 to 0.40. Across all denoising noise levels, distribution-level metrics improve notably—particularly CMMD scores, which decrease consistently: at $\sigma = 15$ from 0.68 to 0.56, at $\sigma = 25$ from 0.72 to 0.65, and at $\sigma = 50$ from 0.73 to 0.67. While CLIP-IQA scores remain relatively stable across configurations, the consistent improvements in FID and CMMD validate that adapting both visual and textual representations enhances the model’s ability to understand and execute restoration instructions.

Figure 2 provides visual evidence of these improvements. The progression from zero-shot to LoRA adaptation, and further to joint text encoder fine-tuning, produces outputs with enhanced structural coherence and more natural appearance. Details are better preserved, artifacts are more effectively removed, and the overall visual fidelity improves substantially. Based on these findings, we adopt joint fine-tuning of text encoders and LoRA adapters for all subsequent

原文截图: Edit2Restore: 用 few-shot LoRA 把编辑模型改造成复原模型, Figure 2, restoration comparison; 来源 PDF 第 9 页。

创新点与真正解决的问题 | Contribution

把自然语言任务提示与 few-shot PEFT 结合，验证编辑模型内部的结构与语义先验可以被快速迁移到多任务复原。

它真正解决的问题是：降低新退化类型的适配数据量和训练成本。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

论文正文报告 1024² 单 H100 约 15–20 s，仍不适合端侧；零样本结果显示大模型先验不能替代任务适配。

适合作为云端按需修复器或新机型快速适配工具；对于端侧需蒸馏成专用小模型。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 生成质量/复原 profile。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

LucidFlux：无 caption 的大规模真实图像复原

论文角色	FLUX.1 backbone
作者/机构	Song Fei; Tian Ye; Lujia Wang; Lei Zhu / HKUST (Guangzhou) / HKUST
发表状态	arXiv:2509.22414
纳入等级	复原重点案例

研究问题与前序缺陷 | Research question

真实照片复原需要理解退化和内容，但 VLM caption 往往把 degradation words 写进条件，造成模型复制或强化噪声。

方法拆解 | Method

LucidFlux 以 FLUX.1 为大规模 diffusion transformer, 构造 caption-free conditioning, 并通过合成/真实混合数据、退化感知训练和视觉质量筛选学习恢复映射。

关键公式 | Key equation: $v_{\theta}(z_t, t, c_{img})$; $L = L_{flow} + \lambda_{rec} L_{rec} + \lambda_{perc} L_{perc}$

Published as a conference paper at ICLR 2026

Semantic Alignment. Preserving semantic fidelity during image restoration is a significant challenge. Existing methods often rely on generating captions from degraded images via vision-language models at inference time (Yu et al. (2024); Ai et al. (2024); Kong et al. (2025)), which causes additional computational cost and may produce inconsistencies between training and inference. Moreover, such captions can explicitly encode degradation-related terms, introducing semantic bias from artifacts rather than underlying content (see Appendix Sec. A.1). Alternative strategies employ coarse textual cues (Wu et al. (2024b;a)), but such signals are generally insufficient to capture fine-grained semantic content. In contrast, LucidFlux leverages a caption-free SigLIP-based semantic alignment module that extracts features from lightly restored proxies and projects them through a lightweight Connector into the text-conditioning space expected by Flux.1, facilitating caption-free guidance without additional VLM calls and avoiding degradation-related caption bias, and ensuring that restored outputs maintain high semantic consistency without hallucinations.

3 METHODOLOGY

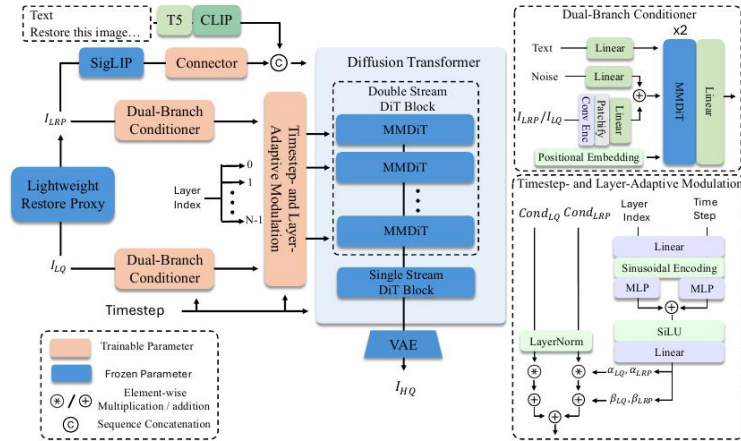


Figure 2: Overview of the proposed architecture for image restoration. Our method integrates dual condition streams (LQ and LRP) with timestep- and layer-adaptive modulation modules, and incorporates SigLIP semantic priors through a connector into a Flux-based DiT backbone to jointly enhance perceptual quality and semantic consistency.

Our framework is built upon a Flux-based DiT backbone, augmented with two parallel ControlNet branches. The first branch processes the original low-quality image (LQ), while the second branch takes a lightly restored version of the input (LRP) generated by a lightweight restoration model. Both streams capture complementary information, which is subsequently modulated through timestep- and layer-adaptive modules to align with the DiT feature space. Moreover, we incorporate semantic priors extracted from SigLIP and enhanced with a connector, which are injected into the DiT layers to facilitate semantic consistency and fine-grained texture restoration.

Practically, we eschew inference-time captions: Appendix Sec. A.1 quantifies that 17–24% of VLM captions introduce degradation-related terms, and Appendix Fig. 5 shows such bias misguides restoration, degrading perceptual quality.

3.1 LIGHTWEIGHT DUAL-BRANCH CONDITIONER

Directly conditioning on the low-quality (LQ) image preserves high-frequency details but often leaks residual artifacts under mixed degradations; conditioning on a lightly restored proxy (LRP)



Figure 4: Qualitative comparisons on RealLQ250. Baseline methods either leave noticeable artifacts or yield over-smoothed textures, while our approach restores sharper details. See Figure 9 to Figure 14 in Appendix for more visual comparisons.

Quantitative Comparisons. Table 1 reports the IQA metric results on real-world and synthetic benchmarks. Our method consistently outperforms prior approaches on perceptual and semantic-oriented metrics, such as CLIP-IQA+, MUSIQ, MANIQA, Q-Align, and NIMA, highlighting its ability to generate visually faithful and semantically aligned restorations. On real-world datasets (e.g., DRealSR, RealSR, RealLQ250), LucidFlux achieves clear gains over existing caption- or tag-based methods. For distortion-focused measures like PSNR and SSIM on synthetic datasets, prior approaches report slightly higher values, yet these metrics are widely recognized as being less correlated with human perceptual quality. In contrast, our method delivers state-of-the-art performance on modern IQA benchmarks, supporting the view that advanced IR frameworks should be evaluated with perceptual and semantic quality measures rather than traditional distortion metrics.

Runtime and Model Scale Comparison. We compare LucidFlux with SeeSR, SUPIR, and DreamClear in terms of runtime and model size in Table 3. Despite using a substantially larger backbone (12B), our LucidFlux achieves a competitive total runtime by eliminating the caption preprocessing. In contrast, SeeSR, SUPIR, and DreamClear require additional preprocessing and rely on smaller backbones (1.29B, 3.5B, 0.6B), resulting in higher latency relative to their size. For trainable adapters, LucidFlux maintains a balanced design (1.6B), outperforming SUPIR (1.3B) in representational capacity while remaining more efficient than DreamClear (2.2B).

Comparison with Close-Source Commercial Methods. We further compare our LucidFlux with several widely used commercial image restoration solutions, including HYPIR-FLUX (Group (2025)), Seedream 4.0 (ByteDance Seed Vision Team (2025)), Topaz (Labs (2025)), Gemini-NanoBanana (DeepMind (2025)), and MeiTu SR (MeiTu (2025)). For clarity, these baselines consist of (i) specialized SR/restoration products (HYPIR-FLUX, Topaz, MeiTu SR) that users rely on in practice, and (ii) unified generative or multimodal systems (Seedream 4.0, Gemini-NanoBanana) that also expose restoration/upscaling interfaces and can handle the same degraded inputs. We include the latter not as dedicated IR baselines, but to contextualize LucidFlux against the current generation of widely deployed unified commercial models. All evaluations are conducted under the same experimental settings, and the identical IQA metrics are used as in the open-source comparisons. Table 2 reports the quantitative results of different methods. Our LucidFlux achieves the

原文截图: LucidFlux: 无 caption 的大规模真实图像复原, Figure 4, RealLQ250 comparison; 来源 PDF 第 9 页。

创新点与真正解决的问题 | Contribution

识别并规避 caption-induced degradation bias: 恢复条件主要来自图像与退化结构, 而不是可能污染语义的文字描述。

它真正解决的问题是: 改善真实低质图像中的去噪、去模糊和细节恢复, 并降低文本条件错误解释的风险。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

caption-free 并不等于 degradation-free；真实数据覆盖、评价指标和高分辨率算力仍需场景化复核。

手机相机 pipeline 更适合图像条件而非自然语言条件；可将 LucidFlux 思路用于 RAW/JPEG/ISP 统计特征的无 caption 适配。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 生成质量/复原 profile。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

ResFlow-Tuner：把推理时计算预算变成 restoration quality 控制量

论文角色	FLUX.1-dev development backbone
作者/机构	Purui Bai; Junxian Duan; Pin Wang; Jinhua Hao; Ming Sun; Chao Zhou; Huaibo Huang / Institute of Automation, Chinese Academy of Sciences / Kuaishou Tech
发表状态	arXiv:2603.22027
纳入等级	复原重点案例

研究问题与前序缺陷 | Research question

真实复原中的输入退化和内容难度差异很大，固定步数/固定预算会导致简单图像浪费计算、困难图像细节不足。

方法拆解 | Method

ResFlow-Tuner 在 inference time 进行 test-time scaling, 通过多候选 flow trajectories、质量排序/选择或调整计算预算, 在不更新主体参数的情况下提升复杂退化恢复。

关键公式 | Key equation: $y^* = \operatorname{argmax}_{\{y_i \in C(x)\}} R(y_i, x); C(x) = \{\text{FlowSolve}(x; B_i)\}_{i=1}^K$

4 Authors Suppressed Due to Excessive Length

performance of pre-trained models during inference by investing additional computational resources, without modifying the model weights. This set of methods has achieved remarkable success in large language models [62] and is rapidly expanding into the visual generation domain. In the visual domain, the most straightforward TTS method is Best-of-N sampling [41][44][46], which directly selects the sample with the highest reward signal from a large number of randomly generated candidates. Subsequently developed Particle Sampling methods [18][21][22][38] retain high-performing sample branches and prune poorer ones during the denoising process, akin to beam search. However, their deterministic nature, due to the lack of stochasticity in the generative process, limits their applicability within more advanced diffusion and flow-based generative models. Furthermore, existing search methods heavily rely on the quality of the initial candidate particles and fail to actively explore new regions of the solution space. To address these limitations, we introduce a novel test-time extension paradigm, specifically designed for the ODE trajectories of flow models in image restoration. This training-free approach dynamically enhances output quality during inference, establishing a new computational-quality Pareto frontier for large models in low-level vision tasks.

3 Method

3.1 Flow-based Image Restoration Framework

Our goal is to leverage the powerful generative prior of the FLUX model for image restoration, guided by both the LQ image and semantic text cues. As illustrated in Fig. 2, our framework consists of two main components: (1) a condition preparation stage, comprising a lightweight degradation removal network and a multimodal large language model for generating textual descriptions; and (2) a conditional injection mechanism, specifically designed for our MM-DiT-based flow model to guide the image restoration process.

Stage-I: Multi-Modal Condition Preparation

- **Reference Image Generation via SwinIR:** Directly processing the heavily degraded I_{LR} with a Vision-Language Model (VLM) can lead to inaccurate semantic interpretations. To mitigate this, we first employ a pre-trained SwinIR model $\mathcal{G}_{SwinIR}$ to obtain a preliminary enhanced image I_{Ref} :

$$I_{Ref} = \mathcal{G}_{SwinIR}(I_{LR}). \quad (1)$$

- **Semantic Context Extraction via VLM:** The reference image I_{Ref} is fed into a powerful VLM $\mathcal{F}_{VLM}$ (e.g., Qwen-2.5-VL) to generate a descriptive textual caption T . This caption encapsulates the high-level semantic content, serving as a weak textual condition to guide the generative process towards semantically coherent content synthesis.

$$T = \mathcal{F}_{VLM}(I_{Ref}). \quad (2)$$

原文截图: ResFlow-Tuner: 把推理时计算预算变成 restoration quality 控制量, ResFlow-Tuner framework; 来源 PDF 第 4

页。

Table 1: Quantitative comparison with state-of-the-art methods on synthetic benchmark. 'I', 'II', and 'III' denote the datasets with mild, medium, and severe degradations, respectively. The best and second performances are marked in **red** and **blue**, respectively.

Dataset	Metric	Real-ESRGAN	BSRGAN	StableSR	DiffBIR	PASD	SeeSR	DreamClear	OSDNet	SUPR	FaithDiff	PaP-Flow	FlowSR	LP	Ours
DIV2K-Vol-I	PSNR $\uparrow$	26.44	27.63	24.71	24.60	25.31	25.08	23.76	25.13	25.09	24.29	23.82	25.02	22.98	23.11
	SSIM $\uparrow$	0.7737	0.7897	0.7131	0.6595	0.6995	0.6967	0.6574	0.7098	0.7010	0.6668	0.6632	0.7219	0.6932	0.6945
	LPIS $\uparrow$	0.1964	0.2038	0.2383	0.2496	0.2370	0.2383	0.2259	0.2382	0.2139	0.2187	0.2214	0.2433	0.2198	0.2098
	PaQ-2-PQ $\uparrow$	73.85	74.17	74.19	72.81	73.12	73.92	73.86	74.07	74.24	74.76	73.21	74.14	73.82	78.82
	MANQA $\uparrow$	0.5611	0.5743	0.5808	0.5521	0.5749	0.5750	0.5767	0.6049	0.5945	0.5511	0.5828	0.5842	0.7261	0.6261
	MUSIQ $\uparrow$	62.38	61.81	65.55	66.23	64.57	66.48	66.15	66.39	65.49	66.53	65.44	66.24	64.85	75.54
DIV2K-Vol-II	CLIPQA $\uparrow$	0.4649	0.4588	0.5156	0.5407	0.4764	0.5336	0.5478	0.5271	0.5202	0.5432	0.5294	0.5328	0.5422	0.7925
	PSNR $\uparrow$	25.49	26.42	24.26	24.42	24.89	24.65	23.39	25.12	24.42	23.80	23.45	24.79	22.54	22.79
	SSIM $\uparrow$	0.7274	0.7402	0.6771	0.6441	0.6764	0.6734	0.6330	0.6983	0.6703	0.6413	0.6421	0.6983	0.6708	0.6778
	LPIS $\uparrow$	0.2309	0.2465	0.2590	0.2708	0.2562	0.2428	0.2518	0.2499	0.2432	0.2407	0.2398	0.2303	0.2346	0.2255
	PaQ-2-PQ $\uparrow$	73.49	73.26	73.81	72.07	72.91	73.32	73.28	73.67	74.26	74.77	73.05	73.82	73.54	78.84
	MANQA $\uparrow$	0.5499	0.5261	0.5825	0.5827	0.5466	0.5691	0.5667	0.5675	0.6012	0.5919	0.5342	0.5682	0.5621	0.7038
DIV2K-Vol-III	CLIPQA $\uparrow$	0.4719	0.4463	0.5057	0.5266	0.4718	0.5226	0.5295	0.5209	0.5202	0.5460	0.5188	0.5204	0.5342	0.7811
	PSNR $\uparrow$	22.81	23.45	23.34	23.42	22.58	22.58	21.82	24.10	21.90	21.77	20.22	21.53	19.48	19.74
	SSIM $\uparrow$	0.6288	0.6281	0.6277	0.5992	0.5985	0.5944	0.5510	0.6432	0.5611	0.5662	0.5034	0.5536	0.5286	0.5263
	LPIS $\uparrow$	0.3555	0.3462	0.3559	0.3676	0.3646	0.3278	0.3336	0.2978	0.3172	0.3080	0.3222	0.3182	0.3228	0.3047
	PaQ-2-PQ $\uparrow$	71.93	72.09	70.22	74.20	72.68	73.46	72.49	73.49	74.07	75.05	73.12	73.48	73.31	78.62
	MANQA $\uparrow$	0.5057	0.4917	0.4896	0.5776	0.4999	0.5462	0.5515	0.5405	0.5882	0.5686	0.4976	0.5206	0.5223	0.6605
LSDIR-Vol	MUSIQ $\uparrow$	60.11	57.49	62.41	58.86	61.08	65.42	62.59	65.28	65.46	66.28	64.12	64.88	61.51	74.47
	CLIPQA $\uparrow$	0.4637	0.4838	0.4124	0.5154	0.4815	0.5106	0.4914	0.5132	0.5134	0.5275	0.5072	0.5126	0.5223	0.7704
	PSNR $\uparrow$	18.13	18.27	18.11	18.42	18.21	18.03	17.01	19.02	16.95	16.87	15.38	16.82	14.82	15.02
	SSIM $\uparrow$	0.4867	0.4671	0.4508	0.4618	0.4592	0.4564	0.4236	0.4796	0.4890	0.4279	0.3802	0.4044	0.3907	0.3845
	LPIS $\uparrow$	0.3986	0.4378	0.4152	0.4049	0.3883	0.3759	0.3836	0.3742	0.4119	0.3802	0.3954	0.3822	0.3810	0.3721
	FDI $\uparrow$	0.2278	0.2539	0.2159	0.2139	0.1734	0.1966	0.1656	0.1619	0.1838	0.1557	0.1675	0.1582	0.1653	0.1406
RealPhoto60	MANQA $\uparrow$	46.46	41.26	51.25	45.91	22.65	25.81	22.06	22.32	30.01	20.49	22.81	21.18	21.76	18.82
	CLIPQA $\uparrow$	0.4381	0.3829	0.3098	0.4551	0.4041	0.5700	0.4811	0.4630	0.4683	0.5956	0.4878	0.6023	0.5912	0.6658
	MUSIQ $\uparrow$	68.25	65.98	59.37	65.94	71.02	73.00	70.40	70.74	70.98	72.64	71.15	71.68	70.82	74.23
	CLIPQA $\uparrow$	0.6218	0.5648	0.5190	0.6592	0.5729	0.7261	0.6914	0.6679	0.6174	0.7224	0.7152	0.7208	0.7097	0.7389
	PSNR $\uparrow$	18.13	18.27	18.11	18.42	18.21	18.03	17.01	19.02	16.95	16.87	15.38	16.82	14.82	15.02
	SSIM $\uparrow$	0.4867	0.4671	0.4508	0.4618	0.4592	0.4564	0.4236	0.4796	0.4890	0.4279	0.3802	0.4044	0.3907	0.3845
RealSR	LPIS $\uparrow$	0.3986	0.4378	0.4152	0.4049	0.3883	0.3759	0.3836	0.3742	0.4119	0.3802	0.3954	0.3822	0.3810	0.3721
	FDI $\uparrow$	0.2278	0.2539	0.2159	0.2139	0.1734	0.1966	0.1656	0.1619	0.1838	0.1557	0.1675	0.1582	0.1653	0.1406
	MANQA $\uparrow$	46.46	41.26	51.25	45.91	22.65	25.81	22.06	22.32	30.01	20.49	22.81	21.18	21.76	18.82
	CLIPQA $\uparrow$	0.4381	0.3829	0.3098	0.4551	0.4041	0.5700	0.4811	0.4630	0.4683	0.5956	0.4878	0.6023	0.5912	0.6658
	MUSIQ $\uparrow$	68.25	65.98	59.37	65.94	71.02	73.00	70.40	70.74	70.98	72.64	71.15	71.68	70.82	74.23
	CLIPQA $\uparrow$	0.6218	0.5648	0.5190	0.6592	0.5729	0.7261	0.6914	0.6679	0.6174	0.7224	0.7152	0.7208	0.7097	0.7389

Table 2: Quantitative comparison with state-of-the-art methods on real-world benchmark. The best and second performances are marked in **red** and **blue**, respectively.

Dataset	Metric	Real-ESRGAN	BSRGAN	StableSR	DiffBIR	PASD	SeeSR	DreamClear	OSDNet	SUPR	FaithDiff	PaP-Flow	FlowSR	LP	Ours
RealPhoto60	PaQ-2-PQ $\uparrow$	69.04	70.4	70.4	64.40	71.88	69.97	73.77	75.84	73.41	74.42	75.27	72.78	73.82	78.38
	MANQA $\uparrow$	0.5094	0.3759	0.4456	0.6253	0.5290	0.6057	0.6166	0.5989	0.6165	0.6527	0.6176	0.6348	0.6320	0.6425
	MUSIQ $\uparrow$	59.29	45.46	51.29	45.91	61.67	64.53	70.46	70.44	70.26	72.74	70.88	71.82	70.96	74.81
	CLIPQA $\uparrow$	0.4389	0.3397	0.4214	0.4935	0.4786	0.5691	0.5273	0.5320	0.5528	0.5932	0.5795	0.5842	0.5856	0.7861
	MANQA $\uparrow$	0.3656	0.3660	0.3465	0.4182	0.3855	0.5189	0.4337	0.4052	0.4296	0.4903	0.4910	0.5232	0.5308	0.6205
	MUSIQ $\uparrow$	62.06	64.67	61.07	61.74	62.03	69.38	61.31	64.87	66.09	69.77	62.27	68.82	61.96	72.34
RealSR	CLIPQA $\uparrow$	0.4872	0.5329	0.5139	0.6202	0.5043	0.6839	0.6895	0.5729	0.5371	0.6804	0.6822	0.6792	0.6842	0.6937
	MANQA $\uparrow$	0.3423	0.3420	0.3171	0.3801	0.3456	0.4974	0.3676	0.3572	0.4174	0.5647	0.5228	0.5424	0.5386	0.6492
	MUSIQ $\uparrow$	58.37	61.22	56.43	55.14	53.82	67.42	59.83	66.68	64.53	68.92	67.12	68.08	68.26	71.82
	CLIPQA $\uparrow$	0.4847	0.5385	0.5344	0.6005	0.5458	0.7022	0.6620	0.6037	0.5800	0.6848	0.6821	0.6946	0.6884	0.7169
	PSNR $\uparrow$	18.13	18.27	18.11	18.42	18.21	18.03	17.01	19.02	16.95	16.87	15.38	16.82	14.82	15.02
	SSIM $\uparrow$	0.4867	0.4671	0.4508	0.4618	0.4592	0.4564	0.4236	0.4796	0.4890	0.4279	0.3802	0.4044	0.3907	0.3845

training. For real-world benchmarks, we conduct experiments on commonly used RealPhoto60 [60], RealSR [5] and DRealSR [51] datasets.

Implementation Details: We build our method upon FLUX.1 [19] and Qwen2.5-VL [4]. By default, we use FLUX.1-dev to generate images based on given low-quality inputs and Qwen2.5-VL-7B to generate captions based on reference images. Our method utilizes LoRA [15] for fine-tuning the base model with a default rank of 16. The SwinIR model in DiffBIR [26] is used as the lightweight degradation remover. Our model is trained on 512×512 resolution images, with a batch size of 16 and gradient accumulation over 4 batch (effective batch size of 64). We employ the Prodigy optimizer [31] with safeguard warmup and bias correction enabled, setting the weight decay to 0.01. Models are trained for 30,000 iterations. For inference of our model, we employ the sde-dpmsolver++ sampler in FlowDPMSolverMultistepScheduler [34] in SDE process with 50 denoising steps and set guidance scale as 5.5. We set the mutation rate $\beta = 0.3$, with σ_t following the default sde-dpmsolver configurations.

4.2 Comparison with State-of-the-Art Methods

We compare our proposed method against several leading approaches, including: GAN-based methods (*i.e.*, Real-ESRGAN [50] and BSRGAN [61]), Diffusion-

原文截图: ResFlow-Tuner: 把推理时计算预算变成 restoration quality 控制量, Tables 1-2, quantitative comparison; 来源

PDF 第 11 页。

创新点与真正解决的问题 | Contribution

将 test-time compute 从固定超参数变成质量-成本控制旋钮, 连接 flow matching 的轨迹求解和 restoration preference.

它真正解决的问题是: 提高不同退化难度下的适应性, 并让质量与延迟可以按业务预算调节。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

候选数量越多延迟越高；reward/ranking 的可靠性直接决定选择结果，存在对文本、纹理和人脸的偏置风险。

可用于云端相册的分级服务：普通照片少预算，夜景/远景/文字照片多预算；端侧需要轻量 quality estimator。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

第四篇 量化、蒸馏和加速 | Efficiency and deployment

效率论文必须拆成四个层级阅读：weight-level (1.58-bit) 、step-level (HiCache) 、activation/kernel-level (Chipmunk) 和 spatial-level (RALU) 。I-Max 是 resolution-extension, 不应简单归类为 restoration。它们的 speedup 不能直接相加，只有在论文明确做了 stacking experiment 时才可引用组合收益。

1.58-bit FLUX：极低比特权重与高效 kernel

论文角色	直接量化 FLUX
作者/机构	Chenglin Yang; Celong Liu; Xueqing Deng; Dongwon Kim; Xing Mei; Xiaohui Shen; Liang-Chieh Chen / ByteDance / POSTECH
发表状态	arXiv:2412.18653
纳入等级	效率图谱

研究问题与前序缺陷 | Research question

FLUX 的参数规模带来显存、带宽和端侧部署压力，传统 FP16 权重存储不利于低成本推理。

方法拆解 | Method

将 FLUX Transformer 权重压缩至约 1.58-bit 表示，并配套高效 kernel；同时比较视觉质量、GenEval/T2I-CompBench 和 latency/memory。

关键公式 | Key equation: $W_q = \alpha \cdot \text{sign/ternary}(W/\alpha)$; $L = L_{\text{task}} + \lambda \cdot qL_{\text{distill}}$

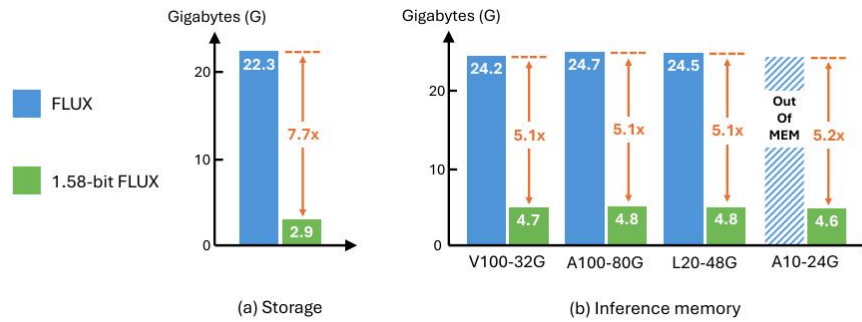


Figure 2. **Efficiency measurements on the vision transformer component of FLUX and 1.58-bit FLUX.** The measurements are based on generating a single image with 50 inference steps. (a) 1.58-bit FLUX reduces checkpoint storage by 7.7 $\times$ compared to FLUX. (b) 1.58-bit FLUX achieves a 5.1 $\times$ reduction in inference memory usage across various GPU types. The x-axis labels, m - n G, represent GPU type m with a maximum memory capacity of n Gigabytes (G).

Abstract

We present 1.58-bit FLUX, the first successful approach to quantizing the state-of-the-art text-to-image generation model, FLUX.1-dev, using 1.58-bit weights (i.e., values in $\{-1, 0, +1\}$) while maintaining comparable performance for generating 1024×1024 images. Notably, our quantization method operates without access to image data, relying solely on self-supervision from the FLUX.1-dev model. Additionally, we develop a custom kernel optimized for 1.58-bit operations, achieving a 7.7 $\times$ reduction in model storage, a 5.1 $\times$ reduction in inference memory, and improved inference latency. Extensive evaluations on the GenEval and T2I Compbench benchmarks demonstrate the effectiveness of 1.58-bit FLUX in maintaining generation quality while significantly enhancing computational efficiency.

1. Introduction

Recent text-to-image (T2I) models, including DALLE 3 [3], Adobe Firefly 3 [1], Stable Diffusion 3 [14], Midjourney v6.1 [41], Ideogram v2 [26], Playground V3 [36], FLUX 1.1 [29], Recraft V3 [2], have demonstrated remarkable generative capabilities, making them highly promising for real-world applications. However, their immense parameter counts, often in the billions, and high memory requirements during inference pose significant challenges for deployment on resource-constrained devices such as mobile platforms.

In this work, we address these challenges by exploring extreme low-bit quantization of T2I models. Among available state-of-the-art models, we select FLUX.1-dev [30] as our quantization target due to its public availability and

competitive performance.¹ Specifically, we quantize the weights of the vision transformer in FLUX to 1.58 bits without relying on mixed-precision schemes or access to image data. The quantization restricts linear layer weights to the values $\{+1, 0, -1\}$, akin to the BitNet b1.58 [40] approach. Unlike BitNet, which involves training large language models from scratch, our method operates as a post-training quantization solution for T2I models.

This quantization reduces model storage by 7.7 $\times$, as the 1.58-bit weights are stored using 2-bit signed integers, compressing them from 16-bit precision. To further enhance inference efficiency, we introduce a custom kernel optimized for low-bit computation. This kernel reduces inference memory usage by over 5.1 $\times$ and improves inference latency, as detailed in Fig. 2 and Tab. 3. Comprehensive evaluations on T2I benchmarks, including GenEval [17] and T2I Compbench [23], reveal that 1.58-bit FLUX achieves comparable performance to full-precision FLUX, as shown in Tab. 1 and Tab. 2.

Our contributions can be summarized as follows:

- We introduce 1.58-bit FLUX, the first quantized model to reduce 99.5% of FLUX vision transformer parameters (11.9B in total) to 1.58 bits without requiring image data, significantly lowering storage requirements.
- We develop an efficient linear kernel optimized for 1.58-bit computation, enabling substantial memory reduction and inference speedup.
- We demonstrate that 1.58-bit FLUX maintains performance comparable to the full-precision FLUX model on challenging T2I benchmarks.

This work represents a significant step forward in making high-quality T2I models practical for deployment on

¹We refer FLUX to FLUX.1-dev in this paper for simplicity.

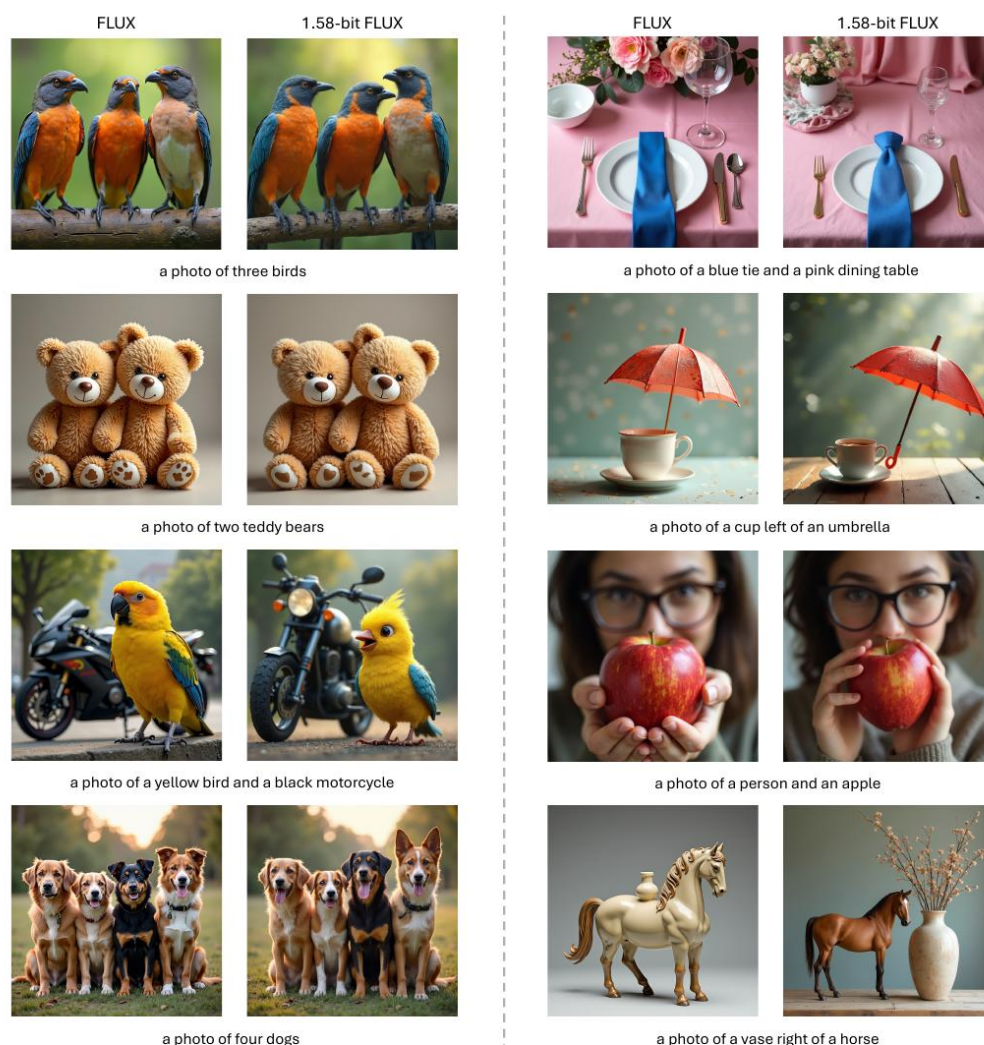


Figure 3. **Visual comparisons between FLUX and 1.58-bit FLUX on GenEval dataset.** 1.58-bit FLUX demonstrates comparable generation quality to FLUX while employing 1.58-bit quantization, where 99.5% of the 11.9B parameters in the vision transformer are constrained to the values +1, -1, or 0. For consistency, all images in each comparison are generated using the same latent noise input. 1.58-bit FLUX utilizes a custom 1.58-bit kernel.

model's total parameters.

Evaluation. We evaluate both FLUX and 1.58-bit FLUX on the GenEval dataset [17] and the validation split of T2I

CompBench [23], following the official image generation pipeline. The GenEval dataset consists of 553 prompts, with four images generated per prompt. The T2I Comp-

原文截图: 1.58-bit FLUX: 极低比特权重与高效 kernel, Figure 3, visual comparison; 来源 PDF 第 4 页。

创新点与真正解决的问题 | Contribution

把低比特权重格式与硬件 kernel 一起评估, 避免只看模型文件大小而忽略解码/矩阵乘的实际收益。

它真正解决的问题是: 降低模型存储、推理显存和部分计算开销。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

量化误差可能集中在 text-image alignment、文字和细节上；kernel 依赖硬件和实现，不应把单卡结果直接外推到手机 NPU。

适合作为端侧部署的模型压缩候选，但必须分别校验 RAW/JPEG restoration 指标和生成型视觉指标。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

I-Max : Projected Flow 扩展 FLUX 的分辨率潜力

论文角色	实验对象 FLUX.1-dev
作者/机构	Ruoyi Du; Dongyang Liu; Le Zhuo; Qi Qin; Hongsheng Li; Zhanyu Ma; Peng Gao / BUPT / Shanghai AI Laboratory / CUHK
发表状态	arXiv:2410.07536
纳入等级	效率图谱

研究问题与前序缺陷 | Research question

预训练 Rectified Flow Transformer 在超出 native resolution 后出现结构坍塌和局部细节不足。

方法拆解 | Method

Projected Flow 在低分辨率 native flow 与高分辨率 latent 之间建立投影/引导，结合 resolution-aware time shift、position index 与 low-resolution guidance。

关键公式 | Key equation: $z_t^H = P(z_t^L) + r_t$; $v_t^H = \text{Proj}(v_t^L) + \Delta v_t$

Technical Report

ation, demonstrating its general applicability to MMDiT structured RFT models. Notably, as shown in Fig. 2, even though Flux.1-dev can generate ultra-high-quality images, conducting resolution extrapolation through I-Max still provides benefits, such as local detail emergence and correcting artifacts like small faces in crowd scenes. Moreover, the overall improvement in generation quality suggests the existence of untapped potential in the model's inference phase. In the field of large language model (LLM) research, there is growing awareness of the potential benefits of increasing inference costs, leading to the concept of the "inference scaling law" [Snell et al., (2024)]. We hope that our work can (to some extent) help similar advancements occur in large vision models.

2 METHODOLOGY

Here, we will introduce the core components of I-Max in the following order: in Sec. 2.1 we first introduce the preliminary knowledge about rectified flow; in Sec. 2.2 we introduce how we achieve low-resolution guidance via projected flow which is tailored for rectified flow models, and in Sec. 2.3 we further summary the inference techniques that are essential for DiTs to infer at extrapolated resolutions.

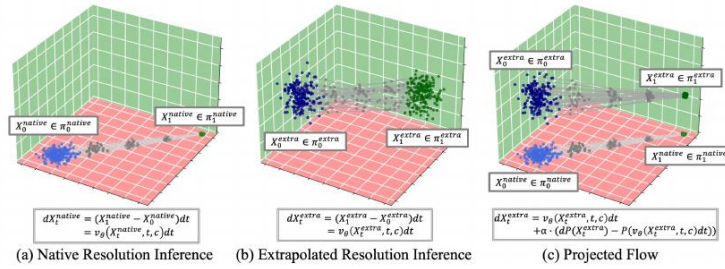


Figure 3: Illustration of Projected Flow Mechanism: (a) In the low-dimensional space of native resolution, RFTs accurately predict the flow direction, ensuring precise distribution transfer. (b) In the high-dimensional space of extrapolated resolution, RFTs struggle to accurately predict the flow direction, degrading the quality of distribution transfer. (c) Projected Flow treats the flow in the low-dimensional space as a deterministic projection of the flow in the high-dimensional space, reducing the difficulty of predicting the flow direction at extrapolated resolutions.

2.1 PRELIMINARY: RECTIFIED FLOW

Rectified Flow [Liu et al., (2022)] and Flow Matching [Lipman et al., (2022)] attempt to simplify the construction of ODE models by linearly interpolating between two distributions. Here, we introduce the preliminary knowledge in the context of Rectified Flow. Given samples $X_0 \in \pi_0$ and $X_1 \in \pi_1$, where π_0 is the noise distribution and π_1 is the image distribution in this paper. Rectified flow builds the path from X_0 to X_1 as a linear flow with the direction of $(X_1 - X_0)$, and the intermediate state X_t can be represented by $X_t = tX_1 + (1 - t)X_0$. Therefore, we can build the ODE of X_t as

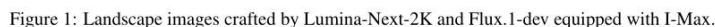
$$dX_t = (X_1 - X_0)dt. \quad (1)$$

Considering that during denoising, the ODE is *non-causal* – since X_1 is the unknown target – we can use $v_\theta(X_t, t, c)$ to fit in the direction $(X_1 - X_0)$ constructing the neural ODE model, where c is additional conditions like text and class labels. In this way, we can transfer X_0 to X_1 by the prediction $d\tilde{X}_t = v_\theta(X_t, t, c)dt$.

2.2 PROJECTED FLOW

Neural ODE models built upon rectified flow (e.g., RFTs) exhibit excellent training and inference efficiency, and they can provide reliable direction predictions $d\tilde{X}_t^{\text{native}} = v_\theta(X_t^{\text{native}}, t, c)$ at the model's native training resolution as illustrated in Fig. 3(a). And when it comes to resolution

原文截图: I-Max: Projected Flow 扩展 FLUX 的分辨率潜力, Figure 3, projected-flow mechanism; 来源 PDF 第 4 页。



Reviewing the existing works on resolution extrapolation, we first decouple the challenges they address into two perspectives: (i) “*how to guide?*”, which refers to how to use reliable native-resolution generation results to guide high-resolution generation, and (ii) “*how to infer?*”, which is about enhancing the model’s generative ability to generalize better to extrapolated resolutions during the inference phase. We consider both perspectives equally essential and build the I-Max framework upon them. However, improving the model’s generalization to extrapolated resolutions seems quite intuitive, while using low-resolution generation for guidance may not appear strictly necessary. For this, we demonstrate through experiments that the diffusion model’s generalization ability to the extrapolated generation resolutions varies over time. It exhibits weaker extrapolation capability during the early denoising stages when generating coarse image content from random noise, but demonstrates a more robust ability during the later stages of denoising the coarse content (please refer to Sec. 3.5). In the following paragraphs, we will summarise how existing works address the questions of “*how to guide?*” and “*how to infer?*”, and introduce our design choices in I-Max that are tailored for RFTs.

不重新训练完整高分辨率模型，而是在 inference trajectory 中引入低维投影和分辨率引导，探索 1K→4K 的生成扩展。

D:\Repository\ReadPaper\daily\Flux

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

它更接近高分辨率生成扩展而非严格 LR-HR restoration；不能把生成的新增细节当成相机真实信息。可作为数字变焦的感知增强分支，但必须与有监督 SR/真实退化模型对照，避免把构图增强误标成恢复。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

Chipmunk : Dynamic Column-Sparse Deltas

论文角色	直接评测 FLUX.1-dev
作者/机构	Austin Silveria; Soham V. Govande; Daniel Y. Fu / UC San Diego / Together AI / Stanford University
发表状态	arXiv:2506.03275
纳入等级	效率图谱

研究问题与前序缺陷 | Research question

FLUX 的跨步激活变化具有稀疏性，但 block sparsity 粒度过粗，动态稀疏模式的索引和 kernel 开销又会抵消收益。

方法拆解 | Method

按列选择变化最快的 activation deltas，结合 voxel-based token ordering、sparse gather、tile packing，并将 pattern identification 与其他计算 overlap。

关键公式 | Key equation: $\Delta h_t = h_t - h_{t-1}$; $S_t = \text{TopK}(|\Delta h_t|)$; $h_t \approx h_{t-1} + M\{S_t\} \Delta h_t$

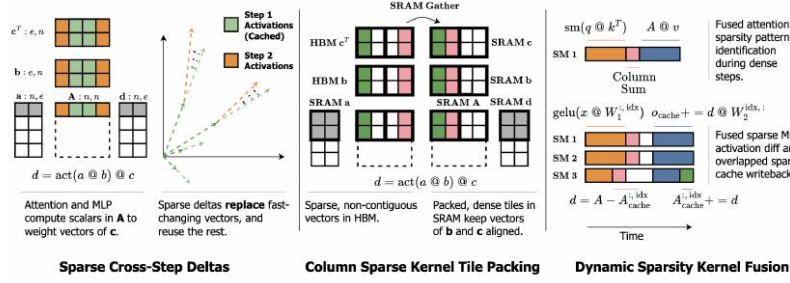


Figure 1: *Left:* Chipmunk uses intermediate activation sparsity to recompute only the fastest changing vectors in the output linear combinations of attention and MLPs. *Middle:* Column sparse kernels achieve low approximation error and hardware-efficiency by packing dense SRAM tiles for peak tensor core utilization. *Right:* Extra operations required to compute dynamic sparse deltas, such as sparsity pattern identification and cache updates, are fused with other operations to minimize overhead.

Existing approaches leverage these redundancies by caching per-step [26], per-layer [50], or per-token [64] outputs, or by computing sparse attention with static local window techniques [56, 61]. In this paper, we ask whether we can achieve greater training-free acceleration by exploiting this redundancy *dynamically*, in the *most granular* manner possible.

We begin with an initial quantitative analysis of activation patterns in two state-of-the-art open source DiTs (HunyuanVideo and FLUX.1-dev). We find that over 90% of the variance in cross-step attention activation changes is explained by only 5-25% of the intermediate activation values (Table 1). That is, recomputing only the top 5-25% of attention interactions and reusing the rest from the previous step captures over 90% of the cross-step change. For MLPs, we find 15-25% of the intermediate activation values explain over 70% of the variance (Table 1).

Chipmunk. This finding motivates our method, Chipmunk, which caches intermediate activations and uses dynamic sparsity to recompute only the fastest-changing activations. We exploit a common computational structure of attention and MLPs— $\text{act}(a \otimes b) \otimes c$ —both using a back-to-back matrix multiply to produce a linear combination of vectors in c (Section 2). Sparsity on the intermediate activations of both operations corresponds one-to-one with these individual vector contributions. Specifically, Chipmunk uses the magnitudes of intermediate activations to determine which vectors of c to recompute at each step, and uses cached values of the rest of the vectors from the previous step. For attention, Chipmunk chooses the top- k *column-chunks* of the attention matrix to recompute, and for MLP, Chipmunk computes an approximate difference against cached activations to identify the top- k column-chunks to recompute. This can be interpreted intuitively as computing sparse deviations from straight line paths in latent space (Fig. 1 left), where the recomputed vectors are dynamically allocated more sequential computation steps.

Systems Challenges. Dynamic sparsity introduces two systems challenges: (1) Sparse attention and MLP operations tend to underutilize tensor cores, which can make it challenging to achieve wall-clock speedup relative to dense baselines. (2) Computing dynamic sparsity patterns at runtime and caching activations introduces additional overhead.

To address these challenges, Chipmunk uses a voxel-based ordering of the pixels (Fig. 2 left) to regroup spatiotemporally local tokens into *column-wise* sparsity patterns (Fig. 1 middle). In particular, a contiguous group of tokens activates the *same* sparse set of individual keys/values (attention) or weights (MLPs). This corresponds to selecting columns of K^T/W_1 and rows of V/W_2 . This column-wise sparsity pattern admits efficient attention and MLP kernels using sparse gathers from GPU global memory (HBM) to packed dense tiles in GPU shared memory (SRAM), which can then fully utilize GPU tensor cores (Fig. 1 middle) [6, 23, 33, 55]. To reduce the overhead of computing dynamic sparsity patterns and maintaining an activation cache, Chipmunk uses custom kernels to overlap sparsity pattern identification and cache I/O operations with other computations (Fig. 1 right). Chipmunk’s column-sparse attention and MLP kernels scale linearly with sparsity with little overhead, achieving nearly-optimal

Method	Efficiency			Visual Quality
	FLOPs ↓	Speedup ↑	Latency (s) ↓	ImRe ↑
FLUX.1-dev, $T=50$ (768 × 1280)				
Flux	100%	1x	6.60s	0.76
STA	84%	1.15x	5.73s	0.75
DiTFastAttn	83%	1.09x	6.05s	0.80
Chipmunk	58%	1.41x	4.90s	0.80
Step+Token Caching (ToCa)	66%	1.51x	4.37s	0.76
Step Caching (TeaCache)	39%	2.51x	2.64s	0.68
Chipmunk+Step Cache	31%	2.56x	2.57s	0.77

Table 5: Performance comparison of various methods for ImageReward for image generation.

Method	Efficiency			Visual Quality	
	FLOPs ↓	Speedup ↑	Latency (s) ↓	GenEval ↑	CLIP ↑
FLUX.1-dev, $T=50$ (768 × 1280)					
Flux	100%	1x	6.60s	0.66	31.07
Step+Token Caching (ToCa)	66%	1.51x	4.37s	0.65	31.21
Step Caching (TeaCache)	45%	2.23x	2.95s	0.61	31.37
Chipmunk-77%+Step Cache	31%	2.56x	2.57s	0.62	31.18
Chipmunk-65%+Step Cache	38%	2.25x	2.93s	0.66	31.43

Table 6: Performance comparison of various methods across various GenEval and CLIP. *Note:* Chipmunk- X denotes a sparsity level of $X\%$ to assess the speed-quality tradeoff.

Baselines. We compare Chipmunk against a number of recent DiT acceleration techniques including TeaCache [26], ToCa [64], Sliding Tile Attention (STA) [61], and DiTFastAttn [56]. TeaCache is a method for dynamically reusing DiT step outputs during slow-changing portions of the generation process, and ToCa dynamically computes token importance scores to recompute activations for only the most important tokens while reusing the rest. STA extends sliding window attention to 3D video to exploit the natural locality in attention computations, and DiTFastAttn reuses non-local attention interactions in a subset of steps to avoid a full recomputation. Here, we also stack Chipmunk with a simple static step cache schedule, untuned for any particular model/prompt. Full baseline descriptions are available in Appendix D.1

Evaluations and Metrics. We evaluate text-to-video generation with the standard VBench evaluation, comprised of 16 dimensions that are weighted to produce a final score shown to align with human judgement [17]. For text-to-image generation, we compute three standard metrics: (i) ImageReward (ImRe) [52], which is a common human-preference trained reward model; (ii) CLIP, which is a widely used metric assessing semantic alignment between text prompts and images (evaluated on a random subset of 10,000 images from MSCOCO2017) [25, 35]; and (iii) GenEval, which is another automatic evaluation metric designed specifically to gauge the quality and accuracy of image generations by comparing them to reference textual prompts, capturing aspects such as relevance, coherence, and descriptive fidelity [14].



Figure 4: Qualitative comparisons across videos (left) and images (right). For videos, frames are stacked vertically (down is later). See Appendix D.2 for more.

原文截图: Chipmunk: Dynamic Column-Sparse Deltas, Table 5 and Figure 4, FLUX evaluation; 来源 PDF 第 9 页。

创新点与真正解决的问题 | Contribution

把稀疏性从 block 级推进到 column 级, 同时围绕 GPU HBM/SRAM/tensor core 数据流设计 kernel。
论文报告 FLUX.1-dev 单独最高 1.41×、与 step caching 叠加 2.25×。

它真正解决的问题是: 在质量基本保持的前提下减少 attention/MLP 的重复计算。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

收益依赖 H100 等 GPU kernel；动态 pattern 的额外开销和稀疏率随分辨率、步数、提示词变化。
云端 GPU 推理可直接受益；手机 NPU 需要另行设计结构化稀疏算子，不能直接复用 H100 speedup。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

HiCache：用 Hermite forecast 改善 FLUX feature caching

论文角色	直接评测 FLUX.1-dev
作者/机构	Liang Feng; Shikang Zheng; Jiacheng Liu; Yuqi Lin; Qinming Zhou; Peiliang Cai; Xinyu Wang; Junjie Chen; Chang Zou; Yue Ma; Linfeng Zhang / Shanghai Jiao Tong University / Fudan University / SCUT / Tsinghua University / HKUST
发表状态	arXiv:2508.16984
纳入等级	效率图谱

研究问题与前序缺陷 | Research question

逐步采样中相邻 feature trajectory 可预测，但 Taylor extrapolation 在 turning points 容易过冲，造成质量损失。

方法拆解 | Method

用 scaled Hermite basis 替换 Taylor predictor，在 cache-then-forecast pipeline 中预测被跳过的 FLUX features，并可叠加 Chipmunk 等稀疏加速。

关键公式 | Key equation: $f(t) \approx \sum_i h_i(\tau) y_i + \sum_i k_i(\tau) m_i$; $R = \text{MSE_Taylor} / \text{MSE_Hermite}$

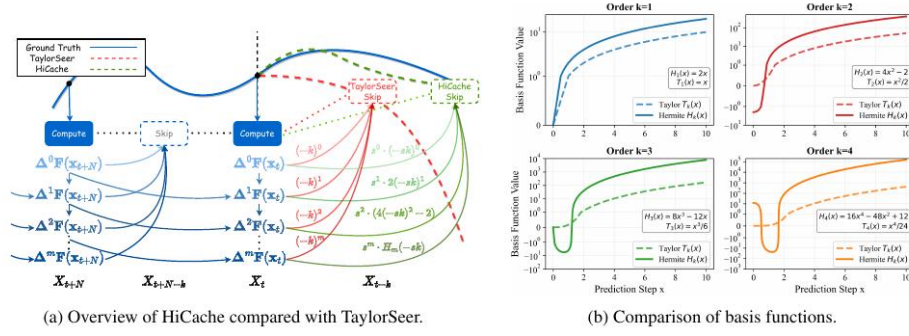


Figure 3: **Method overview and basis function comparison.** (a) TaylorSeer (orange) predicts features using a power basis, whereas HiCache (green) keeps the functional form but swaps the basis for scaled Hermite polynomials. (b) Hermite's oscillatory behavior (e.g., $H_2(x)$ negative offset) captures non-monotonic evolution better than monotonic Taylor growth.

Algorithm 1 HiCache: Hermite-based Feature Caching

Require: Interval N_{interval} , order N_{order} , contraction σ

- 1: Initialize $t_{\text{last}} \leftarrow T$ {Track last activation step}
- 2: **for** timestep $t = T$ to 1 **do**
- 3: **if** $t \bmod N_{\text{interval}} = 0$ **then**
- 4: Compute $\mathcal{F}_t = \Phi_t(x_t)$ {Full model forward}
- 5: Update cache: $\{\Delta^i \mathcal{F}_t\}_{i=0}^{N_{\text{order}}}$ via Def. 1 with $\Delta t_{\text{hist}} = N_{\text{interval}}$
- 6: $t_{\text{last}} \leftarrow t$ {Update last activation step}
- 7: **else**
- 8: $k \leftarrow t_{\text{last}} - t$ {Distance from last activation}
- 9: Predict $\hat{\mathcal{F}}_t$ using Eq. 5 with k
- 10: **end if**
- 11: **end for**

3.3.2 HiCache Algorithm

Proposition 3 (HiCache Feature Prediction). *Given cached derivative approximations $\{\Delta^i \mathcal{F}_t\}_{i=0}^{N_{\text{order}}}$, under Gaussian feature dynamics, a theoretically-motivated predictor is given by:*

$$\hat{\mathcal{F}}_{t-k}^{\text{HiCache}} = \mathcal{F}_t + \sum_{i=1}^{N_{\text{order}}} \frac{\Delta^i \mathcal{F}_t}{i!} \tilde{H}_i(-k) \quad (5)$$

where $k \in \{1, \dots, N_{\text{interval}} - 1\}$ is the prediction horizon and $\tilde{H}_i$ is the scaled Hermite polynomial from Definition 2. When coefficients are obtained via weighted least squares, this achieves projection-optimality in $L^2(\gamma)$. The direct use of $\Delta^i \mathcal{F}_t / i!$ provides an efficient approximation.

HiCache reduces cost by factor $(N_{\text{interval}} - 1) / N_{\text{interval}}$ with $O(N_{\text{order}})$ overhead. The total error $\mathbf{E}_{\text{total}} = \mathbf{E}_{\text{truncation}} + \mathbf{E}_{\text{approximation}} + \mathbf{E}_{\text{numerical}}$ satisfies:

$$\|\mathbf{E}_{\text{total}}\| \leq O\left(\frac{(\sigma\sqrt{2}|\Delta s|)^{N+1}}{\sqrt{(N+1)!}}\right) + O(\Delta t_{\text{hist}}\sqrt{N}) + O(\epsilon_{\text{machine}}) \quad (6)$$

Under $\sigma\sqrt{2}|\Delta s| < 1$, the Hermite truncation $(\sigma\sqrt{2}|\Delta s|)^{N+1} / \sqrt{(N+1)!}$ can be smaller than Taylor's $|\Delta s|^{N+1} / (N+1)!$ due to σ^{N+1} suppression, yielding conditional superiority (see Appendix 4).

原文截图: HiCache: 用 Hermite forecast 改善 FLUX feature caching, Figure 3, Hermite cache forecast; 来源 PDF 第 5 页。

Table 3: Quantitative comparison on the text-to-image generation task with the FLUX.1-dev model. We also report Hi-ClusCa (ClusCa + HiCache) to highlight plug-and-play generality.

Method	Latency(s) ↓	Speed ↑	FLOPs(T) ↓	Speed ↑	Image Reward ↑	PSNR ↑	SSIM ↑	LPIPS ↓
FLUX.1 [dev] - 50 steps [†]	17.12	1.00×	3719.50	1.00×	0.9872	∞	1.0000	0.0000
FLUX.1 [dev] - 25 steps	8.77	1.95×	1859.75	2.00×	0.9691	29.558	0.7327	0.3115
FLUX.1 [dev] - 20 steps	7.07	2.42×	1487.80	2.62×	0.9487	29.122	0.6983	0.3599
FLUX.1 [dev] - 17 steps	6.10	2.81×	1264.63	3.13×	0.9147	28.882	0.6767	0.3912
Chipmunk ^[29] (34% steps)	12.72	2.02×	1505.87	2.47×	0.9936	—	—	—
FORA ($N' = 7$)	4.22	4.08×	670.44	5.55×	0.7418	28.315	0.5870	0.5409
ToCa ($N' = 10, N' = 90\%$)	7.93	2.17×	714.66	5.20×	0.8384	28.761	0.6068	0.4887
TeaCache ($\ell_1 = 1.0$)	4.92	3.48×	743.63	5.00×	0.8379	28.606	0.6360	0.4773
DBCACHE ^[30] ($F = 4, B = 0, W = 4, MC = 10$)	4.08	4.39×	816.65	4.56×	0.8245	—	—	—
ClusCa ($N' = 7$)	4.87	3.52×	674.21	5.52×	0.9480	28.630	0.6210	0.4560
TaylorSeer ($N' = 7, O = 2$)	4.84	3.54×	670.44	5.55×	0.9572	28.634	0.6237	0.4520
Hi-ClusCa [‡] ($N' = 7, O = 2, \sigma = 0.5$)	4.87	3.52×	674.21	5.52×	0.9840	28.940	0.6560	0.4040
HiCache ($N' = 7, O = 2, \sigma = 0.5$)	4.84	3.54×	670.44	5.55×	0.9979	28.937	0.6572	0.3982
FORA ($N' = 9$)	3.90	4.42×	596.07	6.24×	0.5457	28.233	0.5613	0.5860
ToCa ($N' = 12, N' = 90\%$)	7.34	2.34×	644.70	5.77×	0.7155	28.575	0.5677	0.5500
TeaCache ($\ell_1 = 1.2$)	4.45	3.85×	669.27	5.56×	0.7394	28.131	0.4744	0.6765
DBCACHE ^[30] ($F = 1, B = 0, W = 4, MC = 10$)	3.56	5.04×	651.90	5.72×	0.8796	—	—	—
ClusCa ($N' = 9$)	4.53	3.78×	599.93	6.20×	0.8440	28.370	0.5850	0.5170
TaylorSeer ($N' = 9, O = 2$)	4.50	3.80×	596.07	6.24×	0.8562	28.359	0.5882	0.5088
Hi-ClusCa [‡] ($N' = 9, O = 2, \sigma = 0.5$)	4.53	3.78×	599.93	6.20×	0.8860	28.630	0.6380	0.4460
HiCache ($N' = 9, O = 2, \sigma = 0.5$)	4.50	3.80×	596.07	6.24×	0.9113	28.647	0.6443	0.4374
FLUX.1 [schnell] [°] - 2 steps	1.10	15.59×	148.78	25.00×	0.9702	28.053	0.4720	0.6743
FLUX.1 [schnell] - 4 steps	2.20	7.79×	297.56	12.50×	0.9531	28.050	0.4731	0.6708
HiCache+ FLUX.1 [schnell] ($O = 1, \sigma = 0.3$) [*]	1.10	15.58×	148.81	24.99×	0.9975	28.057	0.4507	0.6757

- [†] denotes the baseline (FLUX.1 [dev] - 50 steps). Best results are in **bold** and second best are underlined.
- [‡] Hi-ClusCa: ClusCa with its Taylor predictor replaced by HiCache's Hermite predictor.
- [°] FLUX.1 [schnell] is a distilled variant of FLUX.1.
- ^{*} HiCache+ FLUX.1 [schnell]: Two full computation steps followed by two HiCache-accelerated steps.

shows HiCache's superior temporal consistency compared to methods in terms of both temporal consistency and frame generation.

Class-Conditional Image Generation Table 4 shows that HiCache outperforms recent accelerators on ImageNet with DiT-XL/2 across speedups, with larger relative gains compared with TaylorSeer at higher acceleration—about 3.2% at $\sim 5.0\times$, 4.5% sFID at $\sim 6.2\times$, and 5.9%/6.0% FID/sFID at $\sim 7.1\times$ —while remaining close to the unaccelerated baseline under aggressive acceleration.

Super-Resolution. HiCache achieves $\sim 5.93\times$ theoretical and $\sim 2.43\times$ wall-clock speedup on image super-resolution with Inf-DiT (Yang et al. [2024]), following NTIRE 2025 protocol (Chen et al. [2025]). Figure 7 shows consistent improvements over TaylorSeer in both PSNR and SSIM. At this acceleration, the restoration quality degrades only mildly: PSNR drops by $\approx 2.24\%$ (from 30.87 to 30.18) and SSIM by $\approx 1.44\%$ (from 0.832 to 0.820) relative to the interval= 1 baseline. Detailed latency/FLOPs and restoration metrics for each interval are provided in Table 5.

4.3 Ablation Study

Table 2 dissects our contributions. Without scaling ($\sigma = 1.0$), HiCache underperforms TaylorSeer due to numerical instability. With optimal scaling ($\sigma = 0.5$), performance improves dramatically. The *Hi-Taylor* variant (TaylorSeer with HiCache's dual-scaling) validates that both our scaling mechanism and Hermite basis contribute independently to HiCache's superiority.

5 Discussion

Empirical Validation of Proposition 2 We validate the Gaussianity of finite differences posited in Proposition 2 Using energy tests (Székely and Rizzo [2005]) across 1st to 5th order feature differences in five key FLUX modules (Esser et al. [2024]), the tests confirmed Gaussianity with maximum confidence (p-value = 1.0) across all 25 configurations,

原文截图: HiCache: 用 Hermite forecast 改善 FLUX feature caching, Table 3, FLUX.1-dev comparison; 来源 PDF 第 7 页。

创新点与真正解决的问题 | Contribution

只替换预测基函数, 不新增训练参数; 在 FLUX.1-dev 上报告 $5.55\times$ speedup, 并在 ImageReward 上优于未加速 baseline 的结果。

它真正解决的问题是: 减少重复 denoising block 计算, 同时降低缓存外推的轨迹误差。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

质量结论依赖 prompt、NFE、interval、order 和 GPU；SR 结果来自 Inf-DiT，不能混同为 FLUX-SR 结果。

适合云端批处理和相册精修；端侧需验证缓存的内存占用、热稳定性与动态场景一致性。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

RALU : Training-free Mixed-Resolution Latent Upsampling

论文角色	直接评测 FLUX.1-dev
作者/机构	Wongi Jeong et al. / Seoul National University
发表状态	arXiv:2507.08422
纳入等级	效率图谱

研究问题与前序缺陷 | Research question

DiT token 数随空间分辨率平方增长；直接 latent upsampling 又会产生 aliasing 与 noise-timestep mismatch。

方法拆解 | Method

Region-Adaptive Latent Upsampling (RALU) 只在 artifact-prone edge regions 早期上采样，并通过 noise-timestep matching 对齐不同 latent resolution 的噪声状态。

关键公式 | Key equation: $z^H_t = U_M(z^L_t)$; $t_H = \varphi(t_L, r_H/r_L)$; $M = \text{edge-mask}(z_t)$

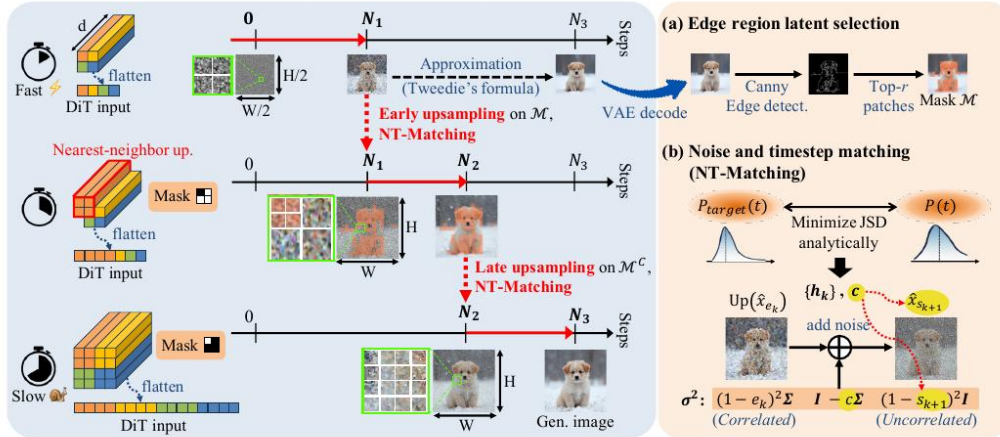


Figure 4. Overview of the proposed RALU framework. RALU consists of three different resolution processes: (1) low-resolution sampling for early denoising, (2) mixed-resolution sampling by upsampling edge region latents, and (3) full-resolution refinement by upsampling all remaining latents. (a) We select the top r fraction of patches with the strongest edge signals from the decoded image and upsample them early. (b) We add correlated noise to the upsampled latents and design a corresponding timestep schedule. (See Sec. 4.2 for more details).

distribution and the modified distribution seems to cause oversampling certain intervals and a frequency imbalance, resulting in visual artifacts. Fig. 3a visualizes this: early upsampling avoids aliasing, but the lack of distribution matching creates global mismatching artifacts.

We found that these artifacts can be mitigated by aligning the upsampling-induced timestep distribution with the original model’s. To verify this, we constructed a two-stage framework with fixed upsampling timestep to $t_{up} = 0.3$ (to prevent aliasing) and applied our corrective noise and timestep distribution (detailed in Sec. 4.2) while varying its parameters to induce a range of Jensen-Shannon divergence (JSD) values. Fig. 3b shows that as JSD decreases, mismatching artifacts diminish (disappearing at $\text{JSD} < 0.03$) and the ImageReward [58] score improves ($p < 0.001$).

Remark 2. Mismatching artifacts in flow-based DiTs occur when naively adding noise after upsampling, but can be mitigated by aligning the new sampling distribution with the original timestep distribution of the pretrained model.

4. Region-Adaptive Latent Upsampling

To break the fundamental trade-off between acceleration and artifacts identified in Sec. 3, we propose Region-Adaptive Latent Upsampling (RALU) approach.

4.1. Mixed-resolution latent upsampling

As illustrated in Fig. 4, our approach comprises mixed-resolution latent upsampling. The denoising process begins

at *low-resolution* to accelerate. We reduce the latent resolution by a factor of 2 along width and height, thereby reducing the number of latent tokens to only 1/4.

Next, we perform an intermediate *edge region upsampling* step to prevent aliasing artifacts, which typically occur in high-frequency regions as discussed in Sec. 3.1. We identify artifact-prone regions by 1) estimating the clean latent $\mathbf{x}_0$ via Tweedie’s formula from the low-resolution stage, 2) decoding it with the VAE, and 3) performing Canny edge detection. We then select and upsample the top- r ratio of latent patches corresponding to these edges. The variable r controls the trade-off between acceleration and artifact and we chose it in the range of 20-30% of the whole latents, making the mixed-resolution latents.

Finally, the remaining low-resolution latent tokens are upsampled to full-resolution to generate a complete high-resolution image. This final step ensures the consistency between the early and late upsampled regions in the final output by matching position embeddings. As demonstrated in Fig. 5, late upsampling (left) results in aliasing, while early upsampling (right) effectively avoids this artifact.

4.2. Noise and timestep matching

As discussed in Sec. 3.2, we design appropriate correlated noise injection and timestep distribution matching after latent upsampling to mitigate mismatching artifacts in DiTs.

Noise injection. Starting from Eq. (3), at the last timestep of the k -th stage e_k , the conditional distribution of the latent after $2\times$ nearest-neighbor upsampling (selected for simplicity;

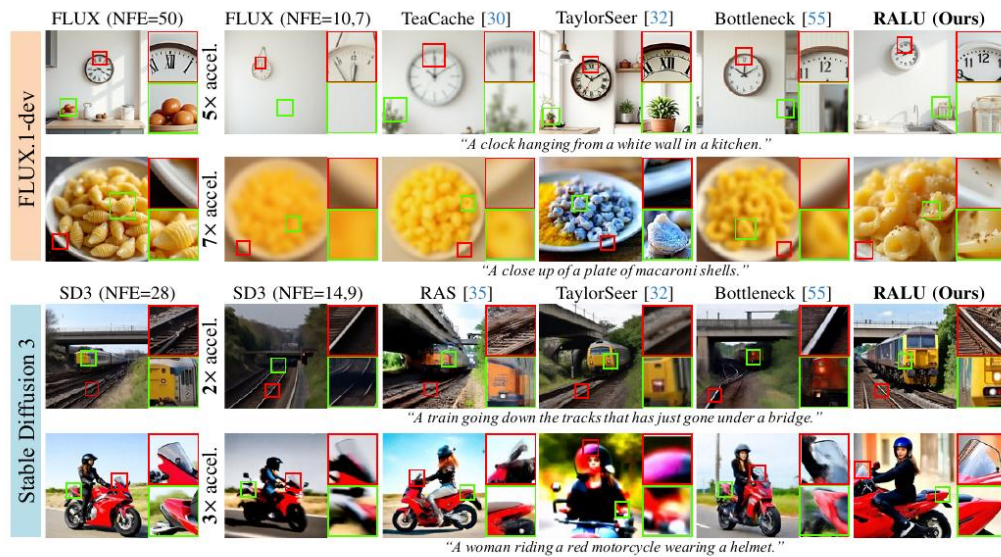


Figure 6. Qualitative comparison of images generated by baseline methods and RALU on FLUX and SD3 under various speedups. For FLUX, we compare at $5\times$ and $7\times$ speedups; for SD3, at $2\times$ and $3\times$. NFE (number of function evaluations) refers to the number of inference steps. Zoomed-in regions on the right highlight that RALU preserves fine-grained details and avoids artifacts more effectively than the other baselines, even under high speedups. More results are provided in the supplementary material. Best viewed in zoom.

gle to deliver strong performance in both image quality and text alignment. The spatial acceleration baseline, Bottleneck Sampling, also suffers from degradation in image quality. In contrast, RALU achieves higher image fidelity and text alignment compared to other baselines.

5.2. Qualitative results

As shown in Fig. 6, base models with reduced inference steps and temporal acceleration methods tend to produce images with noticeable blur or artifacts. Bottleneck Sampling also introduces aliasing or mismatching artifacts. In contrast, RALU outperforms both temporal and spatial acceleration baselines, maintaining superior visual fidelity and semantic alignment with minimal artifacts.

5.3. Integrating RALU with other accelerations

As mentioned in Sec. 1, RALU is complementary to temporal acceleration and can be combined. Tab. 2 (Top) presents the quantitative results of incorporating temporal acceleration methods (e.g., caching, forecasting) into the RALU framework. This integration yields additional improvements in speed up to $9.03\times$ without any training.

Furthermore, RALU is also compatible with timestep-distilled models. Unlike temporal acceleration, which is applied at inference without retraining, timestep distillation

represents a training-based acceleration strategy that fine-tunes the model to operate efficiently in fewer steps. Despite this difference, RALU can be seamlessly integrated with such distilled models to provide further inference-time speedups. As demonstrated in Tab. 2 (Bottom), this combination achieves up to a $15.91\times$ total speedup over the base model with only minimal degradation in generation quality. We also provide qualitative comparisons for these integrations in the supplementary material.

Remark 3. RALU can be integrated with temporal acceleration method or timestep-distilled model to achieve further training-free speedup while incurring only minimal quality degradation.

5.4. Artifact comparison

Fig. 7 shows the artifact ratio according to computational cost (TFLOPs). Bottleneck Sampling [55] suffers from a high artifact ratio, and a naïve upsampling approach also results in a considerable artifact ratio. In contrast, our proposed RALU achieves a substantially lower artifact ratio than both baselines at similar TFLOPs for various speedup settings, demonstrating its superior artifact mitigation.

原文截图: RALU: Training-free Mixed-Resolution Latent Upsampling, Figure 6, accelerated generation comparison; 来源 PDF 第 7 页。

创新点与真正解决的问题 | Contribution

从空间维度减少 token 计算，而不是只跳过时间步；论文在 FLUX.1-dev 上报告最高约 $7.0\times$ ，与 temporal acceleration 组合最高 $15.9\times$ 。

它真正解决的问题是：降低高分辨率 DiT 的空间计算，同时抑制 naive latent upsampling 伪影。

训练、推理与实验解读 | Training and evaluation

- 训练/推理证据：报告只引用正文明确披露的训练目标、采样步数、分辨率或硬件；未披露项标记为未披露，不以经验补齐。
- 评价阅读：PSNR/SSIM 更偏 reconstruction fidelity；LPIPS、CLIP-IQA、ImageReward 和人评更接近 perceptual quality，但不能互相替代。
- 复现边界：arXiv 论文、官方技术报告和模型卡的公开程度不同，代码/权重可用性必须单独核验。

局限、幻觉风险与相机业务启示 | Risk and camera implication

加速结果依赖区域 mask、分辨率和 NFE；它是生成加速，不等同于相机 SR 的 GT fidelity。

对云端 2K/4K 交互预览很有价值，可先低分辨率生成全局结构，再选择边缘区域补算。

在技术谱系中的位置 | Relation

该工作继承了 DiT 的 Transformer scaling、Flow Matching 的速度场学习和 FLUX 的 multimodal latent prior；它将主要创新放在 采样轨迹或推理计算。阅读时应把论文报告的生成能力、恢复能力和工程效率分开比较。

加速层级与适用场景 | Deployment map

层级	代表方法	减少什么	论文证据	端侧注意事项
权重	1.58-bit FLUX	存储/带宽/显存	低比特模型 + kernel	NPU 支持与精度敏感层
时间	HiCache	重复 denoising block	FLUX.1-dev 5.55× speedup	缓存内存与错误累积
激活 /kernel	Chipmunk	attention/MLP 重算	FLUX.1-dev 1.41×；叠加 2.25×	硬件相关，稀疏格式
空间	RALU	高分辨率 token 数	FLUX.1-dev 最高约 7×	mask、边缘和时序匹配
分辨率	I-Max	超 native resolution 失真	1K→4K generation	不是严格 GT restoration

第五篇 综合洞察 | Synthesis

1. 为什么 FLUX 适合真实图像超分？

第一，FLUX 的大规模 image-text pretraining 带来强 semantic prior，能够补全自然图像中的物体结构和纹理关系。第二，flow matching 以 velocity field 描述连续变化，便于使用 trajectory distillation、cache forecasting 和 projected flow 等方法改变计算预算。第三，MMDiT/dual-stream 结构允许图像条件、文本条件和任务控制信号保持清晰的注入路径。

但‘适合’不等于‘天然可信’。相机 SR 的输入只包含有限观测，生成模型新增的纹理可能是 plausible hallucination。实际产品必须区分 enhancement mode 与 faithful restoration mode，并在文字、人脸、车牌、医学纹理等区域使用检测、回退或审计机制。

2. MM-DiT、Flow Matching、VAE、RoPE 与条件注入

- MM-DiT / dual-stream：保留 image tokens 与 text tokens 的模态特定投影，再通过 attention 交互，适合将 LR image、prompt、context image 分别建模。
- Flow Matching：学习 velocity field，而不是只学习 noise residual；它让路径设计和数值求解成为可优化对象。
- VAE：决定像素信息如何压缩到 latent。SR 中 VAE 过强压缩或高频损失会把恢复问题转成生成问题；OP4KSR 通过 F16 VAE 试图缓解 4K token/latent 约束。
- RoPE：将位置信息注入 token attention。分辨率外推会使频率分布失配，OP4KSR 的 RFR 说明高分辨率伪影可能来自 positional frequency，而不只是模型容量不足。
- 条件注入：LR latent、context image、T5/CLIP embedding、ControlNet、LoRA 和 preference adapter 分别控制内容、语义、空间结构、参数更新和恢复意图。

3. 手机相机与云端业务可行性

场景	推荐路线	主要收益	必须补的验证
云端相册精修	FluxSR / DreamSR + HiCache	高质量、可多预算	并发、成本、用户偏好
手机数字变焦	OP4KSR / RALU	无缝 2K/4K、低延迟	ISP 退化、文字/人脸真实性
证件/文档	FoA-SR Faithful	结构保真优先	OCR、几何、不可幻觉

夜景/复杂退化	FluxIR / LucidFlux + gate	强先验补细节	真实噪声、色偏、失败回退
端侧部署	1.58-bit + distillation + sparse	降显存/带宽/功耗	NPU kernel、热、续航、长尾

4. 研究空白与后续阅读路线

- 缺少统一的 FLUX-SR benchmark：不同论文使用的 FLUX 版本、分辨率、NFE、退化模型和指标差异很大。
- 缺少真实相机 provenance evaluation：需要 RAW/JPEG/ISP、镜头、曝光、ISO、运动和多帧信息可追溯的数据。
- 缺少 hallucination audit：PSNR/LPIPS 不能充分检测文字、纹理和身份变化，应加入 OCR、face identity、edge/geometric consistency。
- 缺少端侧证据：H100 的 speedup 不能直接映射到手机 NPU；需要 kernel、内存峰值、温升、功耗和持续运行测试。
- 推荐学习顺序：先读 DiT、Rectified Flow、Flow Matching，再读 SD3/MMDiT 和 Kontext；随后按 FluxSR→DreamSR→OP4KSR→FoA-SR，最后读效率层。

附录 A 全量论文图谱 | Complete paper map

Key	论文	类别	FLUX 角色	本地 PDF
dit	Scalable Diffusion Models with Transformers	01_foundations	theory	dit_2212.09748.pdf
rectified_flow	Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow	01_foundations	theory	rectified_flow_2209.03003.pdf
flow_matching	Flow Matching for Generative Modeling	01_foundations	theory	flow_matching_2210.02747.pdf
ot_cfm	Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport	01_foundations	theory	ot_cfm_2302.00482.pdf
sd3_mmdit	Scaling Rectified Flow Transformers for High-Resolution Image Synthesis	01_foundations	architecture	sd3_mmdit_2403.03206.pdf
flux_kontext	FLUX.1 Kontext: Flow Matching for In-Context Image Generation and Editing in Latent Space	02_flux_foundation	official_report	flux_kontext_2506.15742.pdf
fluxsr	One Diffusion Step to Real-World Super-Resolution via Flow Trajectory Distillation	03_super_resolution	flux_teacher_and_backbone	fluxsr_2502.01993.pdf
dreamsr	DreamSR: Towards Ultra-High-Resolution Image	03_super_resolution	flux_backbone	dreamsr_2605.15682.pdf

	Super-Resolution via a Receptive-Field Enhanced Diffusion Transformer			
op4ksr	OP4KSR: One-Step Patch-Free 4K Super-Resolution with Periodic Artifact Suppression	03_super_resolution	flux_backbone	op4ksr_2605.13457.pdf
foa_sr	FoA-SR: Faithful or Aesthetic? Profile-Aware Preference Optimization for Real-World Image Super-Resolution	03_super_resolution	flux2_backbone	foa_sr_2606.10275.pdf
acquire_adapt_fluxir	Acquire and then Adapt: Squeezing out Text-to-Image Model for Image Restoration	04_image_restoration	flux_data_generator_and_backbone	acquire_adapt_fluxir_2504.15159.pdf
lucidflux	LucidFlux: Caption-Free Photo-Realistic Image Restoration via a Large-Scale Diffusion Transformer	04_image_restoration	flux1_backbone	lucidflux_2509.22414.pdf
resflow_tuner	Tuning Real-World Image Restoration at Inference: A Test-Time Scaling Paradigm for Flow Matching Models	04_image_restoration	flux1_dev_backbone	resflow_tuner_2603.22027.pdf
edit2restore	Edit2Restore: Few-Shot Image Restoration via Parameter-Efficient Adaptation of Pre-trained	04_image_restoration	pretrained_editing_model	edit2restore_2601.03391.pdf

Editing Models

bitnet_flux	1.58-bit FLUX	05_efficiency	quantization	bitnet_flux_2412.18653.pdf
imax	I-Max: Maximize the Resolution Potential of Pre-trained Rectified Flow Transformers with Projected Flow	05_efficiency	resolution_extension	imax_2410.07536.pdf
chipmunk	Chipmunk: Training-Free Acceleration of Diffusion Transformers with Dynamic Column-Sparse Deltas	05_efficiency	sparse_acceleration	chipmunk_2506.03275.pdf
hicache	HiCache: A Plug-in Scaled-Hermite Upgrade for Taylor-Style Cache-then-Forecast Diffusion Acceleration	05_efficiency	feature_cache	hicache_2508.16984.pdf
mixed_resolution_latent	Training-free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers	05_efficiency	spatial_acceleration	mixed_resolution_latent_2507.08422.pdf

注：本附录把理论支撑与正文明确使用 FLUX 的工作放在同一张图谱中，但通过 ‘FLUX 角色’ 区分证据等级。正式发布前可继续追加 OpenAlex/官方页面发现的新论文，并复用本报告的证据规则。

附录 B 原文截图索引 | Figure provenance

- dit | Figure 3, DiT architecture | PDF page 3 | dit_2212.09748.pdf
- dit | Figure 1, generated samples | PDF page 1 | dit_2212.09748.pdf
- rectified_flow | Figure 2, rectified-flow construction | PDF page 4 | rectified_flow_2209.03003.pdf

flow_matching | Figures 2-3, OT and diffusion probability paths | PDF page 6 |

flow_matching_2210.02747.pdf

ot_cfm | Conditional flow matching and minibatch OT | PDF page 3 | ot_cfm_2302.00482.pdf

sd3_mmdit | Figure 2, MMDiT architecture | PDF page 5 | sd3_mmdit_2403.03206.pdf

sd3_mmdit | Figure 1, high-resolution samples | PDF page 1 | sd3_mmdit_2403.03206.pdf

flux_kontext | Figure 4, FLUX.1 Kontext overview | PDF page 5 | flux_kontext_2506.15742.pdf

flux_kontext | Figure 2, iterative editing | PDF page 3 | flux_kontext_2506.15742.pdf

fluxsr | Figure 3, FluxSR training framework | PDF page 5 | fluxsr_2502.01993.pdf

fluxsr | Figure 5, Real-ISR comparison | PDF page 8 | fluxsr_2502.01993.pdf

dreamsr | Figure 2, DreamSR architecture | PDF page 3 | dreamsr_2605.15682.pdf

dreamsr | Figure 4, qualitative comparison | PDF page 7 | dreamsr_2605.15682.pdf

op4ksr | Figure 4, OP4KSR architecture | PDF page 5 | op4ksr_2605.13457.pdf

op4ksr | Figure 8, qualitative comparison | PDF page 11 | op4ksr_2605.13457.pdf

op4ksr | Figure 10, periodic-artifact ablation | PDF page 14 | op4ksr_2605.13457.pdf

foa_sr | Figure 2, FoA-SR framework | PDF page 4 | foa_sr_2606.10275.pdf

foa_sr | Figure 3, faithful/aesthetic comparison | PDF page 8 | foa_sr_2606.10275.pdf

acquire_adapt_fluxir | Figure 3, FluxIR pipeline | PDF page 4 | acquire_adapt_fluxir_2504.15159.pdf

acquire_adapt_fluxir | Figure 5, RealLQ250 comparison | PDF page 7 |

acquire_adapt_fluxir_2504.15159.pdf

edit2restore | Figure 1, few-shot adaptation setting | PDF page 5 | edit2restore_2601.03391.pdf

edit2restore | Figure 2, restoration comparison | PDF page 9 | edit2restore_2601.03391.pdf

lucidflux | Figure 2, LucidFlux architecture | PDF page 4 | lucidflux_2509.22414.pdf

lucidflux | Figure 4, RealLQ250 comparison | PDF page 9 | lucidflux_2509.22414.pdf

resflow_tuner | ResFlow-Tuner framework | PDF page 4 | resflow_tuner_2603.22027.pdf

resflow_tuner | Tables 1-2, quantitative comparison | PDF page 11 | resflow_tuner_2603.22027.pdf

bitnet_flux | Figure 2, memory and latency | PDF page 2 | bitnet_flux_2412.18653.pdf

bitnet_flux | Figure 3, visual comparison | PDF page 4 | bitnet_flux_2412.18653.pdf

imax | Figure 3, projected-flow mechanism | PDF page 4 | imax_2410.07536.pdf

imax | Figure 1, high-resolution generation | PDF page 2 | imax_2410.07536.pdf

chipmunk | Figure 1, dynamic column-sparse deltas | PDF page 2 | chipmunk_2506.03275.pdf

chipmunk | Table 5 and Figure 4, FLUX evaluation | PDF page 9 | chipmunk_2506.03275.pdf

hicache | Figure 3, Hermite cache forecast | PDF page 5 | hicache_2508.16984.pdf

hicache | Table 3, FLUX.1-dev comparison | PDF page 7 | hicache_2508.16984.pdf

mixed_resolution_latent | Figure 4, RALU framework | PDF page 4 |
mixed_resolution_latent_2507.08422.pdf

mixed_resolution_latent | Figure 6, accelerated generation comparison | PDF page 7 |
mixed_resolution_latent_2507.08422.pdf